

Behaviour

A textbook on Behaviour

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Behavioural Science

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Behavioural Science

Definition and Scope of Behavioural Science

Behavioural science is the systematic, evidence-based study of what organisms do — their actions, reactions, and interactions with their environments and with one another — together with the processes that generate, sustain, and alter those actions. This subsection establishes what the term denotes, how the field is demarcated from neighbouring disciplines, and what falls inside and outside its scope. Later subsections examine the history, biological bases, cognitive processes, learning mechanisms, and social influences that the field studies; the purpose here is to fix the object of study and the boundaries of the enterprise.

Defining Behaviour as the Object of Study

Any science is defined in the first instance by its dependent variable. For behavioural science, that variable is *behaviour*: activity of an organism that is in principle observable, measurable, and extended in time. A rat pressing a lever, a shopper abandoning an online basket, a parliament voting on a carbon tax, and a flock of starlings turning as one are all behaviours in this sense — they are events, not states of mind, and they can be counted, timed, coded, or otherwise operationalised.

Two clarifications matter.

Behaviour is not the same as mind. Behavioural science has historically been accused of denying inner life, and some early programmes did restrict themselves to observable responses (a position examined in the Foundations of Behavioural Psychology subsection). Contemporary behavioural science is not eliminativist: it treats thoughts, feelings, and intentions as legitimate causes, provided they are operationalised well enough to yield measurable predictions. Self-report scales, reaction times, physiological indices, and choice data are all admissible, because all are behavioural measures in the broad sense.

Behaviour is not the same as behaviourism. *Behaviourism* is one theoretical tradition within behavioural science — a family of positions about how behaviour should be explained. *Behavioural science* is the wider interdisciplinary field. A cognitive neuroscientist imaging reward anticipation and a sociologist coding protest repertoires are both behavioural scientists, and neither is a behaviourist.

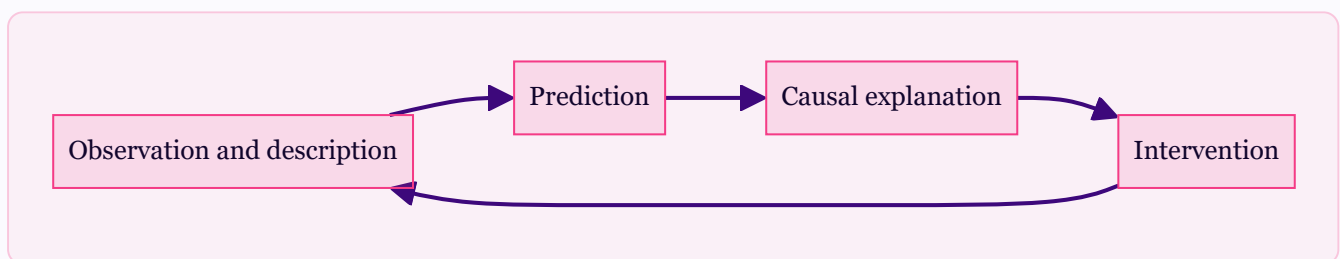
Behaviour is also classified along several dimensions that recur throughout this book:

Dimension	Contrast	Example
Observability	overt vs. covert	Walking vs. privately rehearsing an argument
Level	individual vs. dyadic vs. collective	One person voting vs. a crowd dispersing
Timescale	phasic vs. tonic vs. developmental	A startle reflex vs. a habit vs. a career
Function	instrumental vs. expressive	Working for pay vs. laughing at a joke

What Makes the Study of Behaviour a Science?

Behavioural science earns the label "science" by meeting the same demands as any empirical discipline: publicly specified constructs, systematic observation, quantification, replicability, and theories that generate falsifiable predictions. Its distinctive methodological problems — the reactive subject, the difficulty of randomising life histories, the sheer multiplicity of causes — are treated in Research Methods and Applied Behavioural Science.

A useful way to describe the ambitions of the field is the threefold scheme inherited from experimental analysis of behaviour: **description** (what happens, under what conditions, at what rate), **prediction** (what will happen next, given the current state and context), and **control or explanation** (what interventions change it, and why). Each tier presupposes the one before it. This ordering disciplines the field's claims: a model that predicts a choice but cannot be perturbed to change it remains scientifically incomplete.

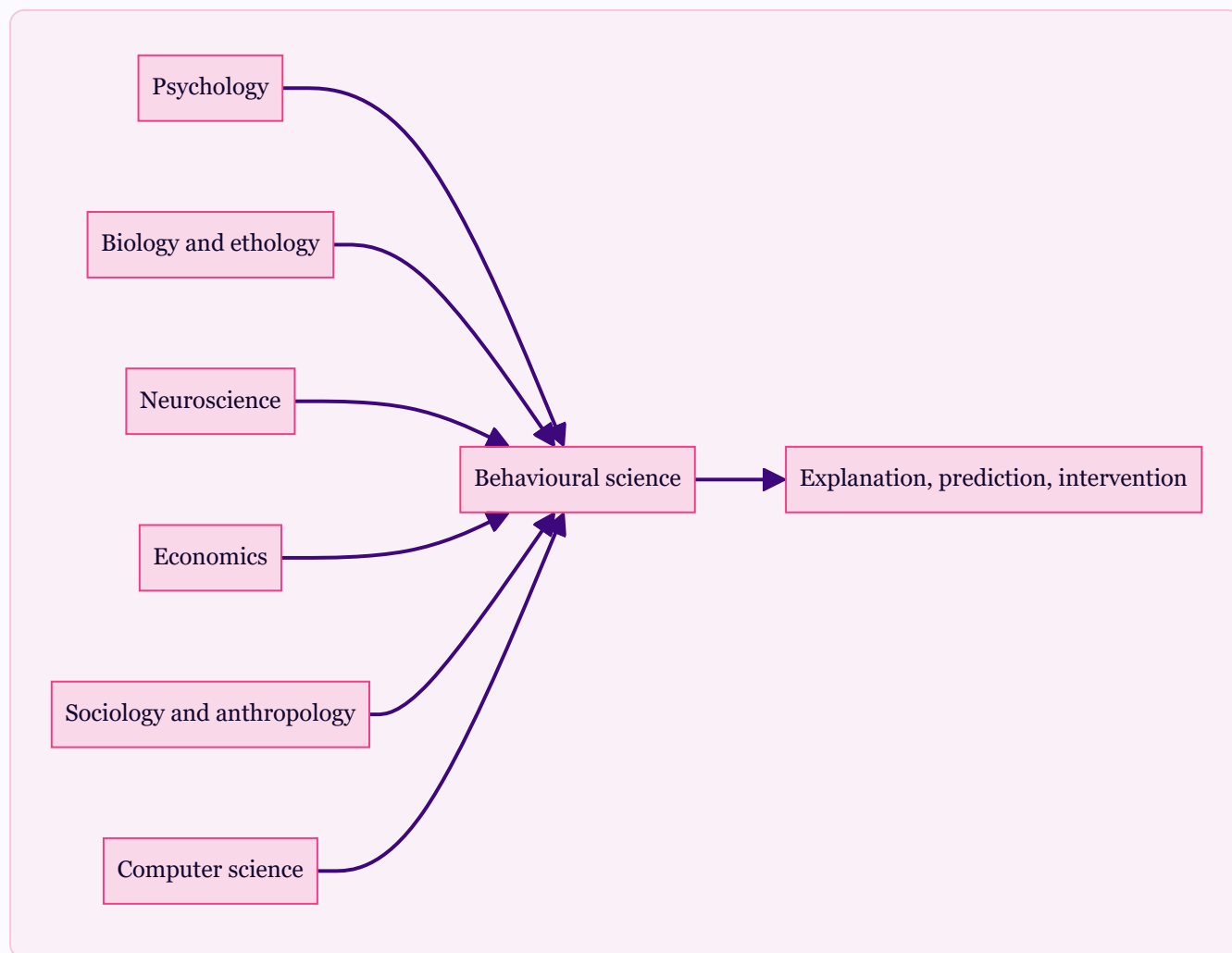


The Disciplinary Composition of Behavioural Science

The field is genuinely interdisciplinary, and the plural form "behavioural sciences" is often used to make this explicit. Its principal contributing traditions include:

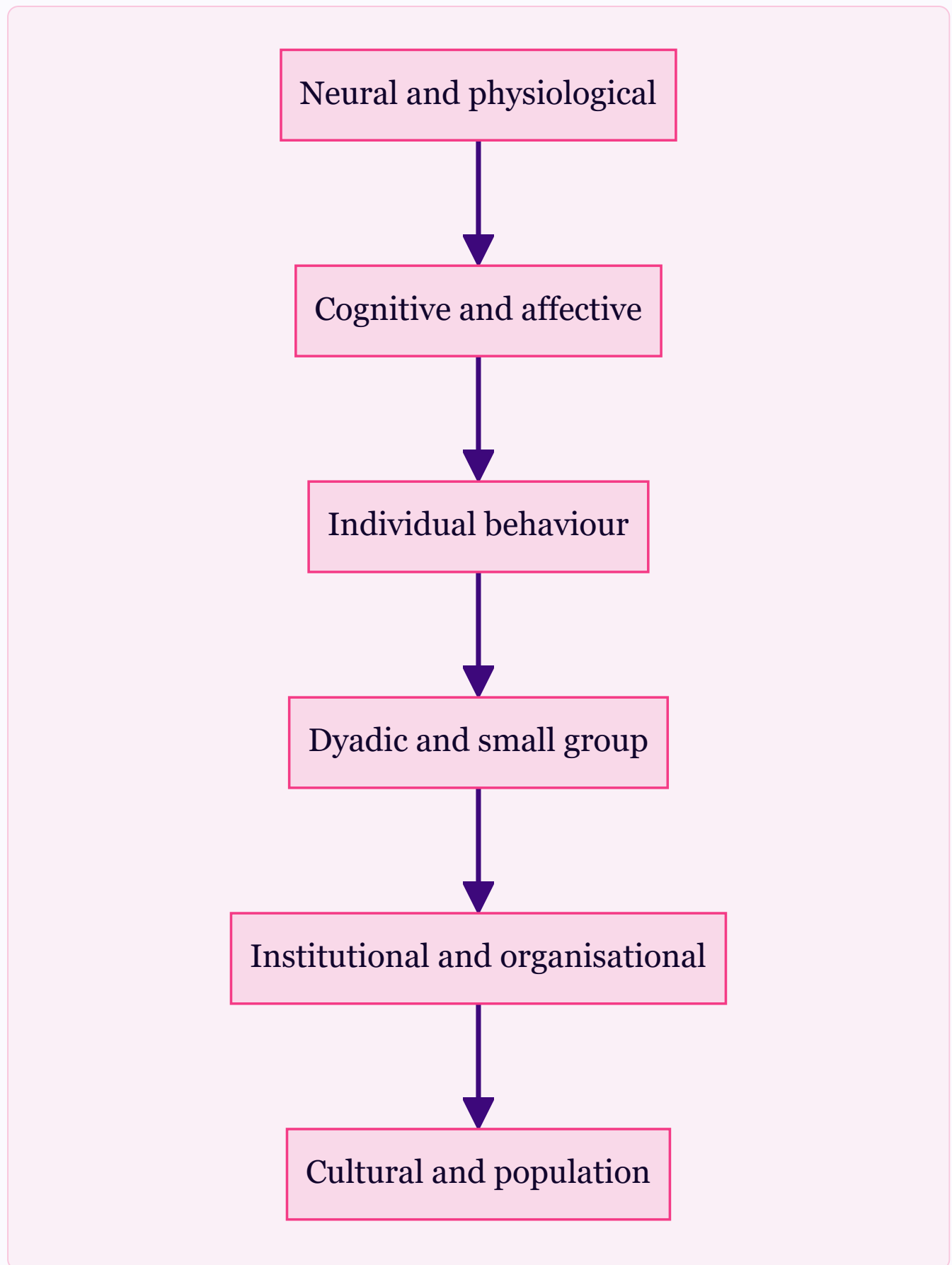
- **Psychology** — experimental, cognitive, developmental, social, clinical, and comparative.
- **Behavioural biology and ethology** — the adaptive function and phylogenetic history of behaviour, including in non-human species.
- **Neuroscience** — the neural mechanisms that implement behaviour.
- **Behavioural economics** — decision-making under bounded rationality, treated in the Behavioural Economics part of this book.
- **Sociology, anthropology, and linguistics** — norms, institutions, culture, and communicative practice.
- **Epidemiology, public health, education, criminology, and computer science** — applied and computational branches, including machine learning of behavioural data.

These traditions converge on the same dependent variable while differing in method, timescale, and explanatory vocabulary.



Levels of Analysis

The single most useful organising idea in the field is that behaviour can be described and explained at multiple levels, none of which is inherently privileged. A standard ascending sequence runs from the molecular and neural, through the cognitive and individual, to the dyadic, group, institutional, and cultural. Explanation at one level does not compete with explanation at another; it answers a different question.



The classic integrative framework for managing these levels is Tinbergen's four questions, which hold that any behaviour is fully explained only when four *separate* questions are answered.

Question	Asks about	Example: prosocial helping
Mechanism	Immediate causal machinery	Empathy circuits, arousal, decision processes
Ontogeny	Development over the lifespan	Parental modelling, socialisation into reciprocity
Function	Adaptive significance	Reputation, reciprocity, group cohesion
Phylogeny	Evolutionary history	Comparative evidence in other primates

Conflating these questions is a common source of sterile debate — for instance, arguing that a neural account of altruism "refutes" an evolutionary one, when the two are answers to different questions.

Worked example: an impulse purchase at the checkout

Consider a shopper who adds a chocolate bar to a grocery order at the last moment. Behavioural science can describe this event at several levels simultaneously:

- **Neural:** cue-triggered dopamine anticipation as the display enters the visual field.
- **Cognitive:** the discounted price serves as an anchor; the item's availability makes it salient.
- **Learning:** previous purchases were reinforced by immediate consumption, building a strong cue-response association.
- **Social:** perceived norms ("everyone buys a little treat"), household roles, and gift-giving conventions.
- **Institutional:** placement at the point of sale, frictionless payment, and credit availability are design choices made by others.
- **Economic:** time-inconsistent preferences and present bias, the subject of the chapter on Present Bias and Self-Control.

No single level is the "real" explanation. A complete account is a coordinated set of answers, and an intervention may target any of them — moving the display, changing the default, or teaching self-regulation.

Core Assumptions and Commitments

Despite their diversity, behavioural scientists share a recognisable set of commitments:

1. **Determinism, weakly stated.** Behaviour is treated as lawful and predictable in principle, even where prediction is practically imperfect.
2. **Empiricism.** Claims are adjudicated by observation, not by intuition or authority alone.
3. **Operationalism.** Abstract constructs must be tied to measurable indicators, or they cannot be tested.
4. **Parsimony.** Simpler explanations are preferred until the data demand otherwise.
5. **Interactionism.** Behaviour is a joint product of person and situation; neither trait nor context alone suffices.
6. **Cumulative theory.** Findings are meant to refine general models, not merely accumulate curiosities.

Contemporary variants add commitments to computational modelling, to dual-process accounts of judgement, and increasingly to prediction and control in real-world settings rather than the laboratory alone.

Scope and Boundaries

The scope of behavioural science can be mapped on three axes: **level of analysis** (neural to cultural), **timescale** (milliseconds to evolutionary), and **domain of application** (health, finance, education, law, environment, technology). The field deliberately spans all three. Boundary questions are nevertheless live:

- **Is subjective experience in scope?** Yes, insofar as it can be measured and shown to do explanatory work — but it is not the field's defining object.
- **Are institutions in scope?** Yes, as causes and consequences of individual behaviour, and as settings where behaviour is aggregated. The sociology of institutions is thus adjacent rather than foreign.
- **Are non-human animals in scope?** Yes; comparative work is foundational to learning theory and to behavioural ecology.
- **Are artificial systems in scope?** Increasingly, where their behaviour is studied with the field's methods, or where they are used as models and as interventions.

By contrast, the field does not claim to settle normative questions. Behavioural science can establish that a nudge changes saving rates; it cannot establish that higher saving rates are desirable. Nor does it subsume the content of economics, law, or ethics — it informs them.

A Brief Note on Applied Scope

The applied reach of behavioural science has expanded sharply since the 2000s through behavioural insights units in governments, the growth of behavioural finance and marketing, the design of digital choice architectures, and public health campaigns. These applications are treated in Behavioural Economics in Policy and Markets and in Behaviour Modification and Applied Extensions. Their prominence has also sharpened ethical scrutiny: interventions that shape behaviour without notice raise questions of autonomy, transparency, and consent, and the field increasingly treats these as part of its subject matter rather than as external constraints.

Limits and Open Questions

Three critical issues define the current frontier of scope. First, **generalisability**: much of the evidence base derives from participants and settings that are unrepresentative of humanity as a whole, which constrains how far findings extend. Second, **replication and effect size**: several headline results in social and behavioural research have proved fragile, prompting methodological reform rather than abandoning the enterprise. Third, **integration**: the field's very breadth makes cumulative theory difficult, and the persistent challenge is to connect levels of explanation without collapsing them into one another. Whether behavioural science is best understood as a single discipline or as a federation of disciplines sharing a dependent variable remains an open question — arguably the field's defining tension.

Summary

- Behavioural science is the systematic study of what organisms do, together with the processes that produce and change it.
- Its object is behaviour: observable, measurable, temporally extended activity. It is not reducible to introspection, and it is not the same thing as behaviourism.
- Its ambitions run from description to prediction to causal explanation and intervention, each tier presupposing the last.

- It is interdisciplinary by constitution, drawing on psychology, biology, neuroscience, economics, sociology, anthropology, and computational fields.
- Levels of analysis run from neural to cultural; Tinbergen's four questions (mechanism, ontogeny, function, phylogeny) discipline how they are combined.
- The field is unified by shared methodological commitments — empirical, operational, parsimonious, interactionist — rather than by a single theory.
- Its boundaries are contested at the edges (institutions, experience, artificial systems) and its scope excludes normative judgement, which it informs but cannot supply.

Historical Foundations and Theoretical Perspectives

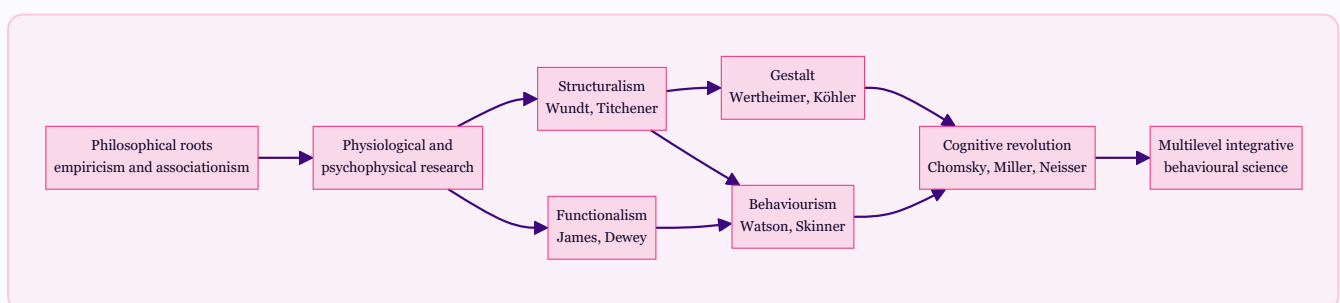
Behavioural science did not arrive as a single, settled programme. It accumulated: a philosophy of mind, a physiology of the nervous system, a psychophysics of sensation, a zoology of instinct, a statistics of individual differences, and a succession of competing "schools" that each claimed the proper unit and method for the study of behaviour. Understanding this history is not antiquarian. The theoretical perspectives available today — behaviourist, cognitive, psychodynamic, humanistic, ethological, ecological, sociocultural, and multilevel-biological — are the surviving descendants of those programmes, and each carries with it assumptions about what counts as an explanation, what counts as evidence, and what may legitimately be ignored.

Philosophical and Pre-Scientific Roots

Systematic reflection on behaviour begins with the philosophy of mind. Descartes's dualism (1637, 1649) separated a thinking substance from an extended body and, in doing so, furnished two legacies that persist in modified form: the *reflex*, a mechanistic account of bodily movement triggered by external stimulation, and the *mind–body problem*, which still structures debates about whether psychological explanation can be reduced to neural explanation.

Against the doctrine of innate ideas, the British empiricists proposed that the mind begins as a blank slate and acquires its contents through experience. Locke (1690), Berkeley (1710), and Hume (1739) supplied the successor principle: **associationism**, in which complex mental contents arise from simple sensations combined by contiguity, similarity, and contrast. Hartley (1749) attempted to tie these associative laws to nervous activity, producing perhaps the first self-consciously *physiological* psychology.

Kant's critique (1781) cut across this programme: if space, time, and causality are contributions of the knowing subject rather than read off from experience, then the empiricist derivation of mind from sensation is incomplete. The resulting **nativism–empiricism** (or nature–nurture) tension has never been resolved; it recurs in every generation, from the instinct debates of the 1920s to contemporary behavioural genetics and gene–environment interaction research.



The Nineteenth-Century Experimental Turn

Three developments converted speculation into measurement.

Physiology. The Bell–Magendie law (1811–1822) distinguished sensory from motor nerve roots, demonstrating that the nervous system is functionally differentiated. Müller's doctrine of specific nerve energies (1833–1840) held that the quality of a sensation depends on which nerve is stimulated rather than on the stimulus itself — a claim that made perception an object of experiment. Helmholtz measured the

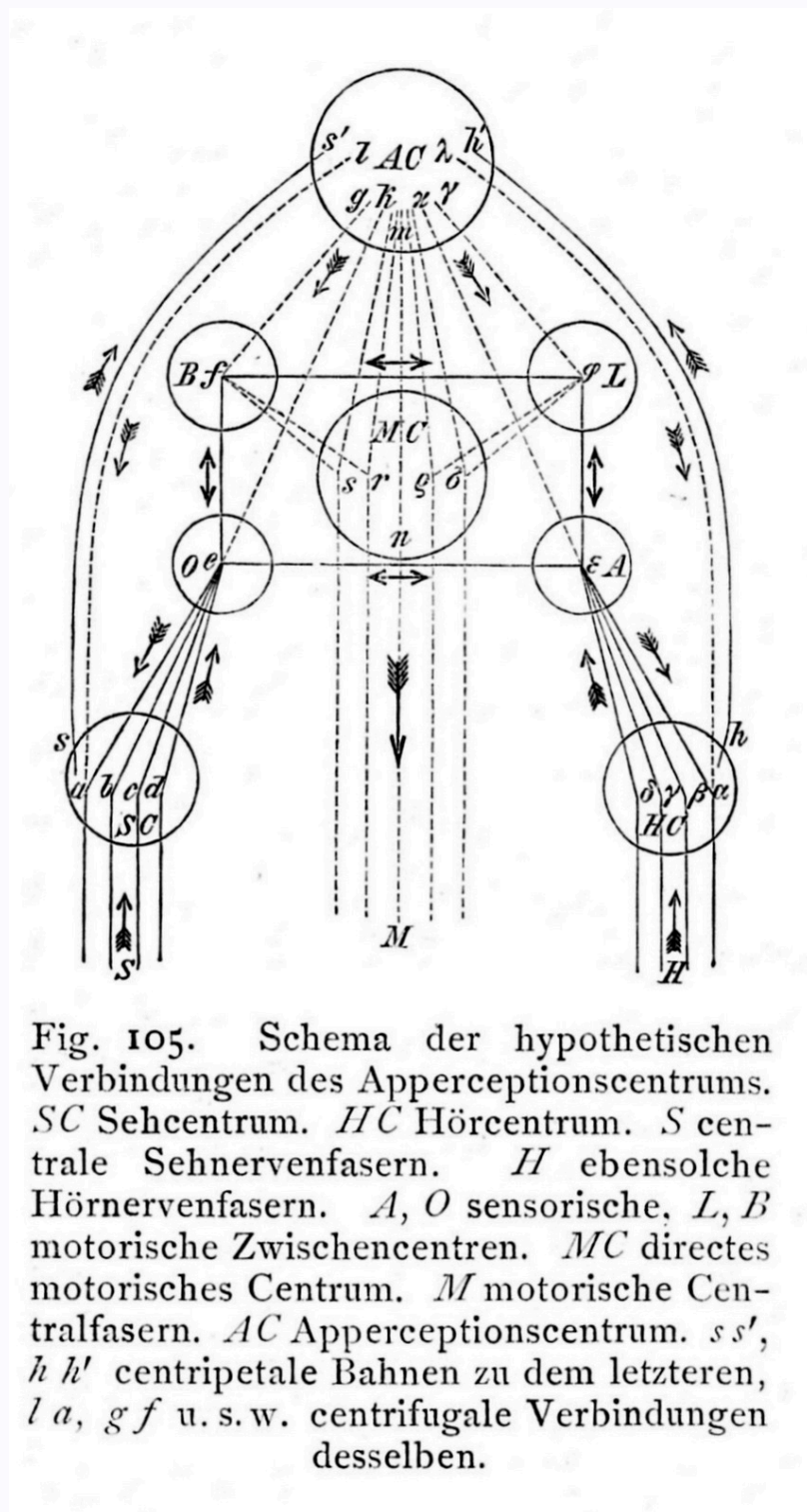
conduction velocity of nerve impulses (1850), refuting the assumption that neural transmission was instantaneous and therefore beyond physical measurement.

Psychophysics. Weber's work on the just-noticeable difference (1834) and Fechner's *Elemente der Psychophysik* (1860) established the first quantitative law relating physical stimulus magnitude to psychological sensation. Donders's subtraction method (1868) used reaction times to infer the duration of mental operations. These were, in effect, the first attempts to treat mental events as measurable intervening variables rather than as metaphysical entities.

Evolution. Darwin's *On the Origin of Species* (1859) and *The Expression of the Emotions in Man and Animals* (1872) made two moves of lasting consequence: they placed human behaviour on a continuum with animal behaviour, licensing comparative study, and they made *function* — what a behaviour is for — an admissible form of explanation. Galton's anthropometric laboratory (1884) and his development of correlation and twin-based reasoning founded the study of individual differences and, with it, the psychometric tradition.

Against that background, Wundt's founding of a laboratory at Leipzig (1879) is conventionally taken as the birth of psychology as an independent experimental discipline. Wundt's programme combined experimental introspection of immediate experience with *Völkerpsychologie*, the study of culturally mediated higher processes. Titchener carried a more austere version to America as **structuralism**, seeking the elementary constituents of consciousness. Structuralism's collapse followed both methodological criticism (the Würzburg controversy over "imageless thought" exposed introspection's unreliability) and theoretical competition.

Functionalism supplied that competition. James's *Principles of Psychology* (1890), Dewey's critique of the reflex arc (1896), and Angell's manifesto (1907) recast psychology as the study of how organisms adapt to their environments, emphasising purpose, habit, and the practical uses of mind. Functionalism was less a doctrine than a set of commitments — Darwinian framing, applied interests, tolerance of methods beyond introspection — and it is the direct ancestor of both behaviourism and modern cognitive-evolutionary work.



Wilhelm Wundt — Public domain, via Wikimedia Commons

The Era of Competing Schools

Between roughly 1910 and 1950, behavioural science was organised into rival systems, each with a founding text, a preferred method, and a claim to the whole field.

- **Behaviourism.** Watson's 1913 manifesto, *Psychology as the Behaviorist Views It*, rejected consciousness as a subject matter and introspection as a method, defining psychology as the prediction and control of observable behaviour. Pavlov's conditioned-reflex work (translated 1927) provided a mechanism; Thorndike's law of effect (1898, 1911) provided a selection principle; Hull (1943) provided formal theory; and Skinner's *The Behavior of Organisms* (1938) provided radical behaviourism, which

reinterpreted "mental" terms as behavioural dispositions. Watson and Rayner's Little Albert study (1920) and Skinner's *Walden Two* (1948) show how far the programme extended into emotion and social design. Its methodological demands — operationism, objective measurement, reproducible preparation — were absorbed by the entire field even as its metaphysics were rejected.

- **Gestalt psychology.** Wertheimer's phi-phenomenon study (1912), Köhler's work on insight in apes (1925), and Koffka's systematic statement (1935) argued that experience is organised into wholes whose properties are not derivable from their parts. The Gestalt demonstrations — perceptual grouping, apparent motion, productive thinking — became core content of cognitive psychology even though the school itself dispersed, partly through the emigration of its founders in the 1930s.
- **Psychoanalysis.** Freud's *Project for a Scientific Psychology* (1895) and *The Interpretation of Dreams* (1900) proposed that behaviour is driven by unconscious conflicts and that early experience leaves durable structural traces. Its influence on clinical practice, developmental theory, and popular conceptions of mind has been immense; its scientific status has been persistently contested, chiefly on grounds of testability and the interpretive latitude of its evidence base.
- **Ethology.** Lorenz's studies of imprinting (1935), Tinbergen's *The Study of Instinct* (1951), and von Frisch's decoding of the bee dance (shared Nobel Prize, 1973) studied behaviour in natural settings, in the animal's own ecology, with an explicitly evolutionary framing. Ethology supplied concepts — fixed action patterns, releasers, sensitive periods — that entered developmental and comparative psychology, and it decisively established that questions about function and phylogeny are scientifically tractable.
- **Humanistic psychology.** Maslow (1943, 1954) and Rogers (1951) positioned their "third force" against both behaviourist mechanism and psychoanalytic determinism, emphasising growth, meaning, and self-determination. Methodologically weaker than its rivals, it nonetheless shaped applied practice, motivational theory, and the study of well-being.

Historians of science caution against treating these schools as clean paradigms in Kuhn's (1962) sense. Their boundaries were porous — Tolman's *Purposive Behavior in Animals and Men* (1932) was behaviourist in method and cognitive in substance — and the "schools" narrative is partly a retrospective tidying by textbook authors.

The Cognitive Revolution and Its Complications

By the mid-1950s the constraints of strict behaviourism were evident: language acquisition, planning, and memory could not be handled convincingly without positing internal representations. The conventional markers are Miller's "The Magical Number Seven, Plus or Minus Two" (1956), Bruner's work on categorisation and the "New Look" in perception, Broadbent's information-processing models (1958), and Neisser's *Cognitive Psychology* (1967). Chomsky's 1959 review of Skinner's *Verbal Behavior* is usually cited as the decisive methodological blow, though its actual impact was slower and more contested than the standard narrative implies.

The cognitive programme was reinforced by older work that behaviourism had marginalised — Bartlett's *Remembering* (1932) on reconstructive memory, Piaget's constructivist developmental theory, Vygotsky's sociocultural account of mediated thought (1934; English translation 1962) — and by two technological currents: the digital computer as a working metaphor for mental processes, and Hebb's *The Organization of Behavior* (1949), which proposed a plausible neural mechanism for learning in cell assemblies. Connectionism (McClelland and Rumelhart, 1986) later softened the computer metaphor into distributed

network models, and the rise of functional neuroimaging in the 1990s gave cognitive claims a biological address.

Biological, Evolutionary, and Ecological Strands

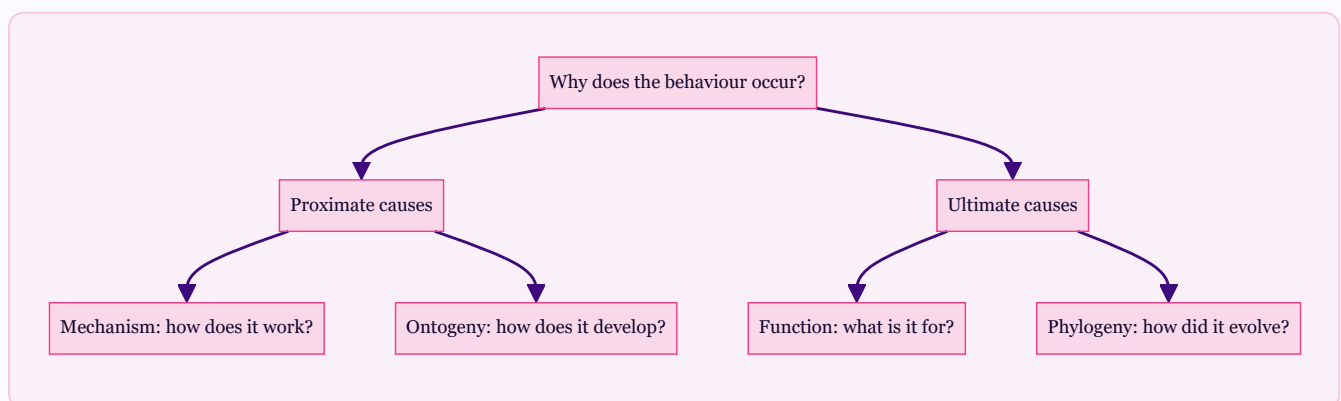
Independently of the cognitive turn, several traditions pressed behaviour into its environmental and evolutionary context.

- **Evolutionary thinking** was formalised by Hamilton's inclusive-fitness theory (1964), Trivers's accounts of reciprocal altruism and parental investment (1971–72), Maynard Smith's evolutionarily stable strategies (1973), and Wilson's *Sociobiology* (1975), whose human applications proved controversial. Evolutionary psychology (Barkow, Cosmides, and Tooby, 1992) recast cognition as a set of domain-specific adaptations shaped in ancestral environments — a productive but vigorously debated research programme.
- **Ecological and systems approaches** relocated the unit of analysis from the organism to the organism–environment system. Gibson's ecological psychology (1979) argued that perception directly picks up *affordances*; Barker's behaviour-setting research and Bronfenbrenner's ecological systems theory (1979) situated behaviour in nested contexts from dyad to culture. Behavioural genetics and gene–environment interaction research completed the picture by demonstrating that nature and nurture are analytically separable but causally entangled.
- **Behavioural economics** imported psychological realism into the study of choice. Simon's bounded rationality and satisficing (1947, 1955) and Kahneman and Tversky's heuristics-and-biases programme (1974) and prospect theory (1979) demonstrated systematic departures from expected-utility predictions. This line of work is treated separately in the behavioural economics chapter, but historically it represents the most successful export of behavioural science into another discipline's core theory.

Integrative Frameworks and Contemporary Pluralism

Two frameworks are now standard for organising this plurality.

Tinbergen's four questions (1963) distinguish proximate from ultimate explanation and, within each, causal-mechanistic from developmental (proximate) and functional from phylogenetic (ultimate) questions. The framework makes clear that competing perspectives are frequently answering *different* questions rather than contradicting one another.



Question	Birdsong example
Mechanism	Syrinx anatomy, song-control nuclei, testosterone
Ontogeny	Sensitive period, tutor model, practice
Function	Territory defence, mate attraction
Phylogeny	Comparative divergence across oscine lineages

Levels of analysis offer a parallel organising device, running from molecular and cellular, through neural circuits and cognitive processes, to behavioural, social, and cultural levels. The biopsychosocial model (Engel, 1977) is its clinical expression. Contemporary behavioural science is therefore pluralistic by design: it assumes that phenomena at one level constrain but do not straightforwardly reduce to phenomena at another, and that a complete account of any behaviour will draw on several levels simultaneously.

This pluralism is disciplined by shared methodological norms rather than shared theory. Meta-analysis, preregistration, and open data have been adopted in response to the replication crisis exposed by large-scale reproducibility projects (notably the 2015 Open Science Collaboration study), and computational modelling and dynamical systems theory (Thelen and Smith, 1994) have supplied formal languages in which behavioural, neural, and developmental claims can be stated precisely enough to be wrong.

Worked Example: Explaining Avoidance of Public Speaking

Consider an adult who reliably declines opportunities to speak in front of groups. Different perspectives do not merely restate the phenomenon in different vocabularies; they generate different predictions and different interventions.

- **Behaviourist:** prior aversive speaking episodes established the situation as a conditioned aversive stimulus, and avoidance is negatively reinforced by anxiety relief. *Prediction:* graded exposure with response prevention reduces avoidance. *Method:* behavioural assessment of approach.
- **Cognitive:** threat appraisals, catastrophising, and attentional bias toward social-evaluative cues maintain the pattern. *Prediction:* cognitive restructuring and attentional retraining change behaviour. *Method:* self-report and reaction-time tasks.
- **Psychodynamic:** the situation activates unconscious conflict over exposure or evaluation rooted in early relationships. *Prediction:* insight-oriented work resolves the pattern. *Method:* clinical case formulation.
- **Evolutionary:** avoidance reflects a mechanism for managing rank and social-evaluative risk in a dominance hierarchy. *Prediction:* the pattern is exaggerated where reputational stakes are high and attenuated where status is secure. *Method:* cross-cultural and comparative comparison.
- **Ecological/systems:** setting features — audience size, seating, evaluation criteria, cultural norms — govern whether avoidance appears. *Prediction:* altering the setting changes behaviour without altering the person. *Method:* ecological momentary assessment across settings.
- **Behavioural-economic:** present bias and loss aversion overweight immediate discomfort relative to delayed reputational gain. *Prediction:* commitment devices and reframing of losses as forgone gains increase participation. *Method:* choice experiments.

A well-designed enquiry does not adjudicate a winner. It specifies which question is being asked, selects the level of analysis at which the answer is expected, and holds the other levels as constraints.

A Caution on Historiography

Two distortions recur. The first is **presentism**: reading earlier programmes as failed attempts at contemporary science, which obscures that behaviourism and structuralism were coherent responses to the epistemic standards of their day. The second is **heroic narrative**: attributing a field's direction to a handful of figures while ignoring institutional factors — the founding of laboratories and journals, the American Psychological Association (1892), wartime applied demands, post-Sputnik funding, and the forced migration of European scholars in the 1930s — that shaped which questions could be asked and funded. The history of behavioural science is better read as a sequence of methodological settlements, each temporarily resolving disputes about what counts as evidence, than as a march toward a single correct theory.

Summary

- Behavioural science emerged from the confluence of philosophical associationism, nineteenth-century physiology and psychophysics, and Darwinian evolutionary thought; Wundt's Leipzig laboratory (1879) marks its institutional origin as an independent discipline.
- Structuralism sought the elements of consciousness through introspection; functionalism, informed by Darwin, redirected attention to adaptation, purpose, and application.
- Between roughly 1910 and 1950 the field was organised into competing schools — behaviourism, Gestalt psychology, psychoanalysis, ethology, and later humanistic psychology — each with distinct assumptions about subject matter and method.
- Behaviourism's operational and experimental standards were absorbed across the field even as its rejection of internal states was abandoned; the cognitive revolution reinstated representation, supported by information-processing models, Hebbian neuroscience, and later connectionism.
- Evolutionary, ecological, and systems perspectives established that behaviour must be analysed in relation to environment, development, and phylogeny, while behavioural economics demonstrated the practical reach of psychological findings into decision theory.
- Tinbergen's four questions and multilevel analysis are the principal contemporary devices for reconciling these perspectives; they reframe apparent disagreements as answers to different questions.
- The historical record should be read with attention to presentist and heroic distortions: theoretical succession reflects changing methodological settlements and institutional conditions, not a linear approach to a single correct account.

Biological and Evolutionary Bases of Behaviour

Behaviour has a body. Every act a science of behaviour records — a lever press, a hesitation at a ballot box, a defensive flinch — is implemented by tissue, and every disposition to produce such acts has a history in populations that extends far past the lifetime of the individual. This subsection surveys the two biological enterprises that follow from that observation: the study of the **proximate machinery** that produces behaviour here and now (genes, neurons, hormones, and the developmental processes that assemble them), and the study of the **evolutionary processes** that shaped why such machinery exists at all. Tinbergen's separation of these questions, introduced earlier, is assumed throughout; the aim here is to examine what each tradition has established and how the two interlock.

The Substrate: Genes, Neurons, and Hormones

From genotype to behaviour

No gene codes for a behaviour. Genes encode proteins, proteins regulate cellular physiology, cells form circuits, and circuits interacting with an environment produce actions. The causal chain is long, conditional, and regulated at every step — transcription, splicing, translation, receptor density, and protein degradation are all sensitive to experience. This is why the language of "genes for aggression" or "a gene for curiosity" is not a harmless shorthand but a category error. What exists are alleles that shift the probability distribution of a behavioural phenotype, usually by a small amount and usually only in particular environments.

Two caveats discipline the literature. First, dramatic single-gene effects on behaviour occur almost entirely in pathological conditions: mutations affecting language development, rare variants producing severe impulsivity, or neurological disorders with clear behavioural signatures. These are informative about mechanism and almost useless as models of normal variation. Second, ordinary behavioural variation is massively **polygenic**: hundreds or thousands of variants, each with a negligible effect, jointly account for a modest and long-debated fraction of phenotypic variance. Larger and larger genome-wide association samples have raised the reliability of polygenic scores without making them anything resembling a person-level predictor.

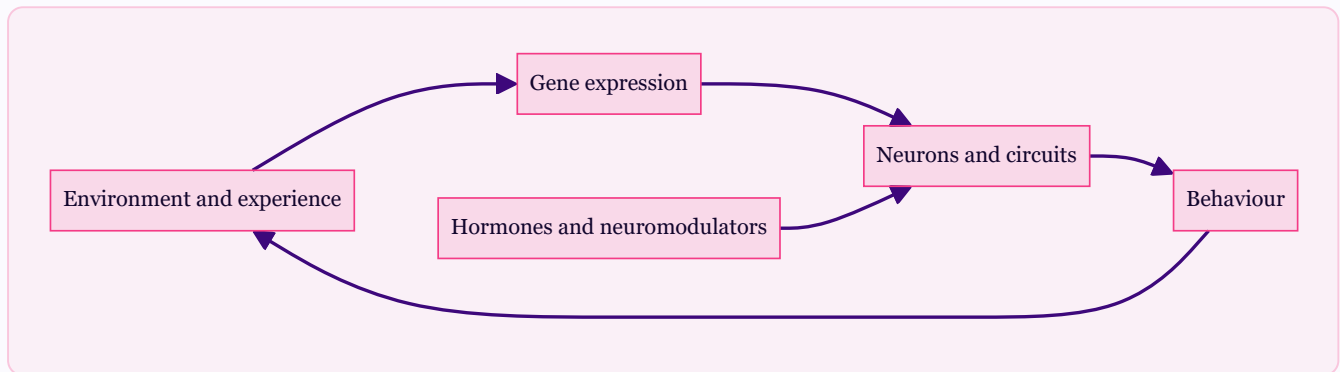
Neurons, circuits, and the fallacy of one-to-one mapping

At the level of the nervous system, behaviour is produced by populations of neurons whose firing patterns are sculpted by synaptic plasticity. Several organising principles are worth fixing.

- **Circuit specialisation with degeneracy.** Distinct systems carry distinct functions — mesolimbic dopamine circuitry supporting reward learning and incentive motivation, amygdala-centred circuitry supporting threat acquisition and expression, prefrontal systems supporting working memory and inhibitory control, hippocampal systems supporting episodic and spatial memory. But the mapping is many-to-many: a single behaviour engages multiple circuits, and a single circuit contributes to many behaviours. Incentive-salience accounts of dopamine, for instance, explain approach and "wanting" better than they explain hedonic pleasure.
- **Neuromodulation.** Diffuse systems — dopaminergic, serotonergic, noradrenergic, cholinergic, oxytocinergic — tune the parameters of local circuits rather than transmitting specific content. Popular summaries routinely oversimplify these: the "serotonin imbalance" account of depression is not

supported by the best current evidence, and oxytocin's effects on affiliation reverse with context, group membership, and personality. Neuromodulators alter gain and salience; they do not carry the message.

- **Hormones.** Hormonal influences are conventionally divided into **organisational** effects, which act during development to shape structure durably, and **activational** effects, which modulate already-differentiated systems in adulthood. Testosterone in birdsong nuclei and in aggression illustrates both: it organises circuits perinatally and activates singing or competitive behaviour seasonally. Importantly, hormonal effects on human social behaviour are bidirectional and modest — a competitive encounter raises testosterone as much as testosterone produces competitiveness — so the "hormone causes behaviour" framing is usually the wrong direction of fit.



The diagram captures the essential circularity: behaviour is an output of biological machinery and simultaneously an input that reshapes it, both within a lifetime (plasticity) and across generations (selection).

Behaviour Genetics and Gene–Environment Interplay

Behaviour genetics estimates the contribution of genetic differences to differences among individuals in a population, chiefly through twin, adoption, and, increasingly, molecular designs. Three well-replicated regularities are worth stating because they are so frequently misread.

1. **Nearly every measured behavioural trait is moderately heritable.** Personality dimensions, cognitive ability, temperament, and even political orientation and voting turnout show substantial twin-study heritability.
2. **Shared environment contributes less than intuition suggests** to adult behavioural variation, though it matters greatly in childhood and for some outcomes throughout life.
3. **A large residual remains unaccounted for** by genes and by family environment as conventionally measured — a residual that includes measurement error, idiosyncratic experience, and non-additive effects.

Heritability is a population statistic, not a property of an individual, and it is not fixed. It varies with environmental range: where environmental conditions are homogenised, genetic differences account for a larger share of the remaining variance, and where environmental deprivation is severe, genetic differences may barely express themselves at all. The frequently repeated claim that a trait is "70 per cent genetic" says nothing about any person's modifiability.

Molecular work has complicated the picture further. Twin studies of height imply heritability around 80 per cent, yet polygenic scores built from enormous samples still fall well short of that figure — the "missing

heritability" problem — and for behavioural traits the shortfall is larger still. Genome-wide polygenic scores for educational attainment, for instance, predict statistically significant but individually negligible variance, which makes them unsuitable for selection or diagnosis and valuable mainly for research.

Gene–environment interplay is the rule rather than the exception and takes several distinguishable forms:

- **Interaction (G×E):** the same genotype has different effects in different environments, and vice versa. The widely cited finding that a monoamine-oxidase variant moderates the effect of childhood maltreatment on antisocial behaviour is the canonical example — and also a canonical example of a result that has proved difficult to replicate, which should temper confidence in any single reported interaction.
- **Correlation (rGE):** genotypes and environments are non-randomly paired. Parents transmit both genes and rearing conditions (**passive**), children evoke responses according to temperament (**evocative**), and adolescents select niches matching their dispositions (**active**). Active and evocative processes mean that environment is partly phenotype, so "environmental" effects are never purely exogenous.
- **Epigenetic regulation:** experience can alter gene expression durably without altering sequence. In rodents, variation in maternal licking and grooming alters methylation of a glucocorticoid-receptor promoter and, through it, stress reactivity in offspring — a concrete mechanism by which early care calibrates a physiological system. Transposing such findings to humans requires caution, but the general principle, that the environment acts through biological channels rather than alongside them, is secure.

Worked example: why the same upbringing produces different adults

Two siblings raised in one household differ markedly in anxiety. A purely environmental account looks for differential treatment; a purely genetic account looks for different alleles. The behaviour-genetic account says both are operating and neither is sufficient:

- **Genetic variation** shifts thresholds for threat reactivity in the two children.
- **Evocative correlation** means the more anxious child receives more protective parenting, amplifying the disposition.
- **Non-shared environmental experience** — different peers, illnesses, teachers — supplies divergence that family-level measures cannot detect.
- **Developmental timing** means the same caregiving has different effects at ages three and thirteen.

The practical implication is that interventions modifying family environment will shift the *distribution* of anxious behaviour without eliminating individual differences, and that a claim that a trait is "genetic" does not license a claim that it is unchangeable.

Development and Plasticity

Brains are not preformed; they are constructed, and construction is activity-dependent. **Long-term potentiation** and its counterpart depression provide the cellular basis for Hebbian learning — the strengthening of synapses that fire together — which underlies much of what the learning literature treats functionally.

Two developmental concepts organise the timing of this construction. **Experience-expectant** processes require a species-typical environmental input during a **sensitive period**, and if the input is absent or

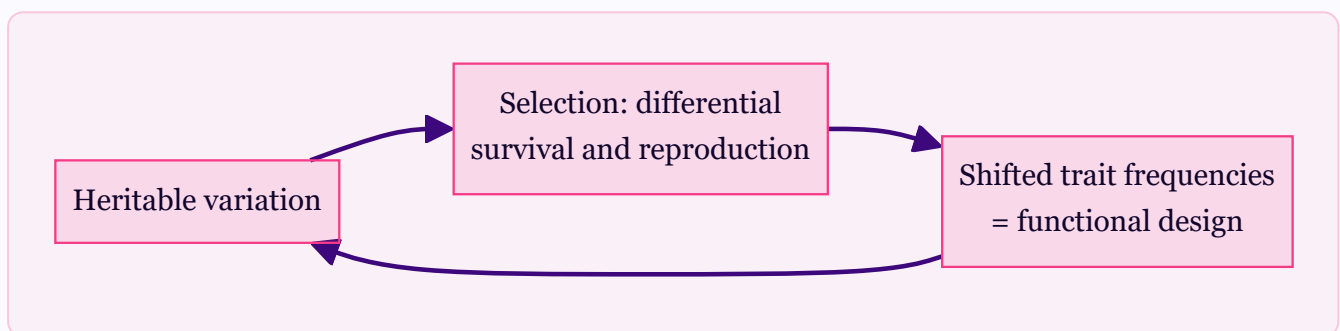
abnormal the system does not develop normally. Visual cortical organisation in early life, and song learning in oscine birds — where a tutor must be heard within a restricted window — are the standard illustrations.

Experience-dependent processes continue throughout life and encode information peculiar to the individual.

Sensitive periods are better understood as windows of heightened plasticity than as hard deadlines; their closure is itself an active, experience-influenced process. This matters for applied work: intervention after a sensitive period is possible, often harder, and frequently more costly in the plasticity it demands.

Evolutionary Foundations

Where the preceding sections ask how behaviour works, evolutionary analysis asks why a behavioural disposition exists and persists. The logic is deceptively simple: if behaviour varies, if that variation is heritable, and if variants differ in their consequences for survival and reproduction, then trait frequencies in the population will change. Repeated over generations, this process produces **design** — organised structure with the appearance of purpose.



Several refinements do most of the explanatory work.

- **Inclusive fitness and kin selection.** Hamilton's rule ($rb > c$) states that a costly act spreads when the benefit to the recipient, discounted by the coefficient of relatedness, exceeds the actor's cost. This dissolves the apparent paradox of altruism toward relatives and explains the extreme reproductive division of labour in eusocial insects.
- **Reciprocity.** Trivers's account of reciprocal altruism shows that cooperation among non-relatives can be stable where interactions are repeated, partners are identifiable, and defection is punishable. Empirical work on reciprocal grooming, coalition formation, and contingency in helping has largely borne this out, while documenting the cognitive demands of tracking exchange.
- **Parental investment and family conflict.** Because parents are equally related to all offspring while offspring are more related to themselves than to siblings, Trivers predicted conflict over weaning and resource allocation. Parent–offspring conflict is now documented across taxa, and is a useful reminder that "family harmony" is not the evolutionary default.
- **Sexual selection.** Competition within a sex, and choice by the other, can drive traits whose costs to survival are substantial. Zahavi's handicap principle and Fisher's runaway process offer different mechanisms by which costly displays become honest signals; both appear to operate in nature.
- **Game theory and evolutionarily stable strategies.** Maynard Smith's framework formalised the analysis of frequency-dependent selection. Hawk–dove and related models show that mixtures of strategies, rather than single optima, are often the stable outcome, and they connect evolutionary biology directly to behavioural economics.

- **Life-history theory.** Organisms allocate finite resources among growth, maintenance, and reproduction. This generates trade-offs — current versus future reproduction, quantity versus quality of offspring — that predict coordinated suites of behavioural and physiological traits across species and, more controversially, across human developmental ecologies.

Adaptation, By-Product, and Constraint

A trait's presence is not proof that selection built it *for* its current effect. Gould and Lewontin's critique of "adaptationist" reasoning insisted that architecture, developmental constraint, and historical accident produce features that are by-products of other adaptations, or that were co-opted for new functions (**exaptation**). Behavioural science therefore needs criteria for adjudicating adaptive hypotheses, and the standard battery includes comparative evidence of convergence across independent lineages, evidence of heritable variation in the trait, demonstration that the trait actually affects fitness in the predicted direction, and experimental manipulation.

Two further constraints deserve mention. **Phylogenetic inertia** means lineages inherit legacies that are not currently optimal; **mismatch** means that mechanisms calibrated to past environments can misfire in present ones — the appetite system calibrated to scarcity operating in a calorie-dense environment is the canonical case. Comparative analysis must also worry about **homology versus analogy**: if two species share a behaviour through common descent, that is not independent evidence of adaptation, which is precisely why phylogenetically controlled comparative methods displaced naive cross-species counting.

Evolutionary Psychology and Its Critics

The programme known as evolutionary psychology treats the mind as a collection of domain-specific mechanisms shaped by selection in ancestral environments. Its productive hypotheses include preparedness in fear learning, cheater-detection in social exchange, sex-differentiated mating preferences, and patterned responses to environmental harshness and unpredictability in development.

The programme has attracted sustained criticism on four fronts. First, adaptive explanations are easy to generate and hard to falsify, so the field's standards of evidence matter more than its ingenuity. Second, the posited ancestral environment is often treated as more uniform and static than the archaeological and ethnographic record warrants. Third, strong claims about massive modularity understate developmental plasticity and domain-general learning, and sit uneasily with the same authors' recognition of cultural variation. Fourth, evidence drawn overwhelmingly from a narrow slice of the world's populations constrains the generality of the conclusions — the generalisability problem discussed in the section on limits and open questions.

The critiques do not license discarding evolutionary reasoning; they demand that it be disciplined. Three developments in particular have moved the field forward:

- **Gene–culture coevolution.** Cultural transmission is itself a selective pressure. Lactase persistence in dairying populations, and the behavioural and physiological changes accompanying domestication, are standard demonstrations that genes and culture evolve jointly rather than in parallel.
- **Niche construction and reciprocal causation.** Organisms modify their environments, and thereby modify the selection pressures acting back on them. Behaviour is not merely the output of evolution; it is a driver of it.

- **Comparative cognition.** The social brain hypothesis relates relative brain size to the demands of social living, and convergent findings in corvids and cetaceans — caching, tool use, perspective-taking — show that similar cognitive solutions arise in independent lineages. Such convergence is among the strongest forms of evidence available for functional hypotheses.

Multilevel selection remains a live dispute: whether adaptations can be attributed to groups, or whether group-beneficial outcomes are always reducible to individual and kin-level processes, is contested, and positions have shifted more than once. The dispute is instructive because it shows that the field's core disagreements are now about the *units and levels* of selection rather than about whether selection explains behaviour at all.

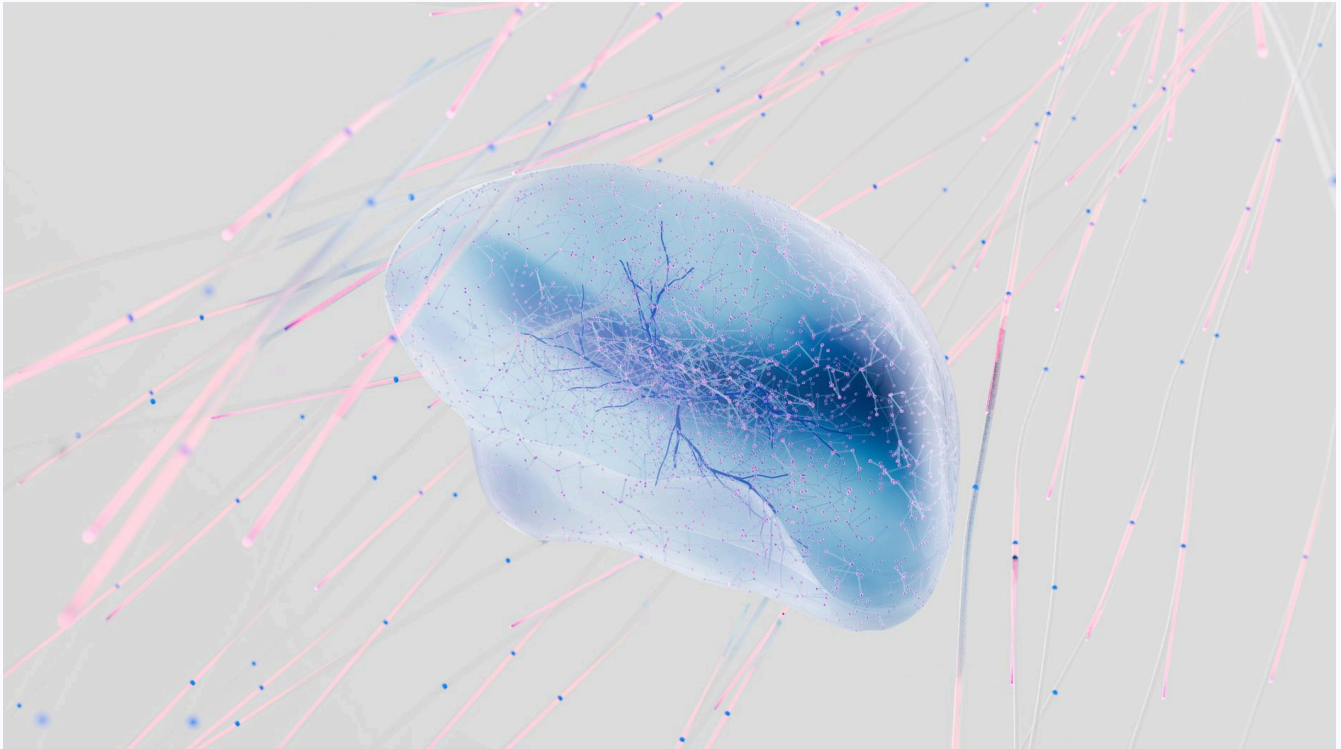


Photo by Google DeepMind on Pexels

Worked example: the fear of snakes and spiders

Why do humans so readily acquire fears of snakes and spiders, and why do such fears resist extinction? No single level answers the question, and each level generates a distinguishable prediction.

- **Phylogenetic:** Serpents were a recurrent, often lethal hazard for ancestral primates. Comparative evidence suggests specialised detection mechanisms in primates broadly, not in humans alone.
- **Functional:** Rapid detection and avoidance of venomous animals carried direct survival benefits, favouring attentional prioritisation over deliberate evaluation.
- **Mechanistic:** Threat stimuli reach the amygdala rapidly, and fear conditioning proceeds through circuits that can operate with minimal cortical involvement. Prepared cues are learned with fewer pairings and extinguished more slowly than arbitrary ones — the **preparedness** effect.
- **Ontogenetic:** Infants show attentional biases toward snakes and spiders, but full-blown fear typically requires social information — witnessing a parent's alarm, or culturally transmitted warnings. Fear is not merely built in; it is scaffolded.
- **Genetic and developmental:** Individual variation in fearfulness is moderately heritable and interacts with caregiving and with unpredictable environments.

- **Mismatch:** The mechanism fires on pictures, garden spiders, and harmless species — the signature of a system calibrated to a different ecology.

The practical payoff is that the levels converge on one implication and diverge on another. They converge in predicting that exposure-based intervention must address a rapid, non-deliberative pathway, which is why fear reduction is often slow. They diverge in predicting *what* to modify: attentional retraining, verbal re-appraisal, community norms, or environmental design. A well-specified account says which of these it is testing.

A cautionary parallel comes from evolutionary accounts of psychopathology. Hypotheses that depression reflects a socially conservative response to rank loss, or rumination as a mechanism for solving intractable problems, are heuristically fertile but notoriously difficult to test against the alternatives, and they risk normalising conditions that have clear proximate causes and effective treatments. Evolutionary reasoning is a source of hypotheses, not a substitute for clinical evidence.

Summary

- Behaviour has a biological substrate; genes influence behaviour only through proteins, cells, and circuits, and only in interaction with environments.
- Ordinary behavioural variation is polygenic and conditional; heritability is a population-level statistic that varies with environmental range and does not license claims about individual modifiability.
- Genes and environments are entangled through interaction, correlation, and epigenetic regulation, so "nature versus nurture" is not an empirical question but a discredited framing.
- Neural and hormonal systems support behaviour with many-to-many mappings; circuit specialisation coexists with degeneracy, and neuromodulators tune processing rather than carrying content.
- Development is constructive and activity-dependent, organised around sensitive periods of heightened plasticity rather than fixed critical deadlines.
- Evolutionary analysis explains design through heritable variation and differential reproduction, extended by inclusive fitness, reciprocity, parental-investment theory, sexual selection, and evolutionary game theory.
- Not every trait is an adaptation; by-products, exaptation, phylogenetic inertia, and environmental mismatch all require consideration, and adaptive hypotheses must be tested with comparative, experimental, and fitness evidence.
- Evolutionary psychology has been productively extended by gene–culture coevolution, niche construction, and comparative cognition, while remaining subject to serious criticisms about testability, assumptions regarding ancestral environments, modularity, and the narrow sampling base of its evidence.

Cognitive, Affective, and Motivational Processes

Behavioural science treats behaviour as its dependent variable, but the prediction of that variable usually requires constructs that sit between stimulus and response. This subsection examines three families of such constructs: **cognitive** processes, which acquire, transform, and deploy information; **affective** processes, which assign value and prepare action; and **motivational** processes, which select, energise, and sustain courses of action. These are psychological-level constructs in the sense set out earlier — they are admissible in a science of behaviour precisely insofar as they are operationalised, measured, and shown to carry predictive or explanatory weight. They are not rivals to neural or social accounts; they answer Tinbergen's question of *mechanism* at a different grain from the synapse or the institution, and the same event can legitimately be described at all three levels at once.

The three families are treated separately below only for exposition. In practice they form a loop: what an organism perceives and believes shapes what it feels; what it feels shapes what it wants; what it wants shapes what it attends to and remembers; and the outcomes of its behaviour feed back into all three.

Cognition: The Processing Architecture of Behaviour

Cognition denotes the set of processes by which information is acquired, transformed, stored, retrieved, and used to guide action. Its defining behavioural signature is that **identical stimuli produce different responses** depending on what the organism knows, expects, or has previously encountered. A clinician reading a chart sees a pattern where a novice sees a list of numbers; the light entering the eye is the same.

Attention is the selection mechanism that makes this possible. Capacity is limited, so processing is biased toward stimuli that are salient, unexpected, or currently goal-relevant. Attention runs in two broad modes: *stimulus-driven* (a sudden movement captures the gaze) and *goal-directed* (a driver scans for a specific street sign). The distinction matters behaviourally because failures of attention, not deficits of knowledge, underlie many everyday errors — the missed instruction, the unnoticed change in a routine form. Sustained attention also degrades predictably over time, a vigilance decrement with direct consequences for inspection, monitoring, and driving tasks.

Memory supplies the accumulated context against which current input is interpreted. Working memory holds and manipulates a small amount of information over seconds; long-term memory divides usefully into episodic (events), semantic (facts), and procedural (skills and habits) stores, of which the last connects most directly to the learning mechanisms treated elsewhere in this part. Three behavioural implications recur throughout applied work: retrieval depends heavily on the match between encoding and test context; memory is *reconstructive* rather than reproductive, so confident recall can be systematically wrong; and prospective memory — remembering to perform a future intention — fails disproportionately under divided attention, which is why well-designed systems externalise intentions rather than relying on them.

Schema-driven processing extends this further. Stored knowledge structures guide perception and inference, filling in missing information and speeding recognition at the cost of rigidity. The same ambiguous behaviour in a meeting may be coded as "decisive" or "abrasive" depending on the observer's prior schema, and once coded as one, the observer tends to notice subsequent evidence that fits.

Executive function names the control processes that coordinate the rest: inhibiting prepotent responses, updating working memory contents, and shifting between tasks or rules. These capacities predict a wide

range of behavioural outcomes — academic persistence, adherence to treatment regimens, financial planning — because they govern the ability to act on delayed rather than immediate contingencies, the topic developed in the chapter on Present Bias and Self-Control.

Finally, **judgement and decision-making** research documents systematic departures from normative models: reliance on availability, representativeness, and anchoring; sensitivity to how options are framed; and asymmetric weighting of losses relative to equivalent gains. These findings are treated in detail in the behavioural economics part of this book; here it suffices to note that they are cognitive phenomena with direct behavioural consequences, and that they are ordinarily *adaptive* heuristics rather than breakdowns.

Affect: Emotion, Mood, and Stress

Affect is the umbrella term for valenced states — states experienced or expressed as pleasant or unpleasant, activating or deactivating. Within it, two constructs are conventionally distinguished. An **emotion** is a relatively brief, multi-componential episode directed at a specific object or event; a **mood** is lower in intensity, longer in duration, and typically diffuse in its object. **Stress** refers to the affective and physiological response to demands that exceed, or are appraised as exceeding, available resources.

The componential view of emotion is now standard: an emotion is not a single thing but a coordinated pattern across several channels, which is why it can be measured in several ways and why those measures frequently dissociate.

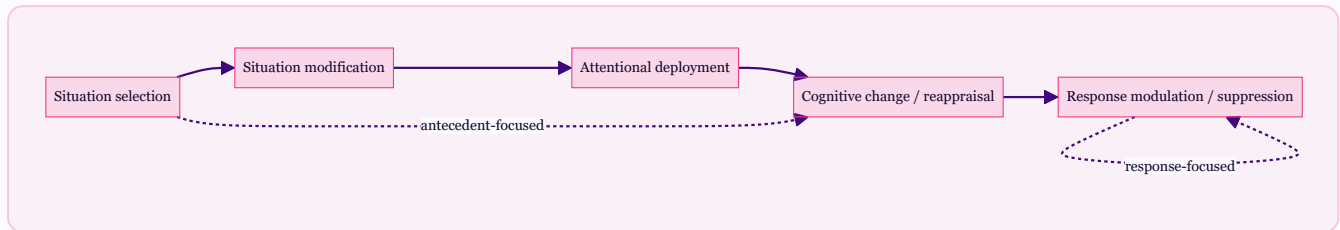
Component	What it is	Typical measure
Appraisal	Evaluation of the event's relevance to goals and coping options	Self-report, reaction time, vignette ratings
Physiological	Autonomic and endocrine changes	Heart rate, skin conductance, cortisol
Expressive	Facial, vocal, postural signals	Facial EMG, coding systems, acoustic analysis
Action tendency	Preparedness for a class of response	Approach/avoidance tasks, behavioural choice
Subjective feeling	The experienced quality of the state	Self-report scales

Appraisal theory is the hinge between cognition and affect. On this account, an emotion is generated not by an event as such but by the person's evaluation of it — its relevance to goals, its implications, the agent responsible, and the perceived capacity to cope. Lazarus distinguished a *primary* appraisal (is this relevant and threatening or beneficial?) from a *secondary* appraisal (can I handle it?). The same examination can therefore produce anxiety or challenge depending on whether resources are judged adequate, and this judgement is itself manipulable by information, reframing, or prior experience.

Affect is not merely a by-product of behaviour; it performs work. It signals value, recruits physiological resources, communicates states to others, and biases subsequent cognition. Two findings illustrate the last of these. **Mood-congruent processing**: people in a positive mood retrieve more positive material and interpret ambiguity more favourably. **Affect as information**: when a judgement is difficult or the basis for it is unclear, people consult how they feel and treat that feeling as evidence — a shortcut that is usually

efficient but easily contaminated by irrelevant sources such as weather or a prior unrelated task. The relationship between arousal and performance is likewise non-monotonic, generally described by an inverted U: too little arousal yields inattention, too much yields disorganisation, particularly for complex tasks.

Emotion regulation — the processes by which people influence which emotions they have, when, and how they express them — is best understood as a sequence of opportunities rather than a single act.



The model's practical value is diagnostic: interventions that act early in the sequence (avoiding or altering the situation, redirecting attention, reappraising meaning) generally carry lower costs than those acting late, because suppression of an already-generated response leaves the physiological and cognitive components intact while adding the effort of concealment. Chronic reliance on late-stage strategies is associated with poorer affective and interpersonal outcomes.

Motivation: What Initiates and Sustains Behaviour

Motivation refers to the processes that select, energise, direct, and sustain behaviour. It is analytically distinct from learning: learning changes the repertoire of what an organism *can* do, while motivation determines what it *does* with that repertoire at a given moment. A person may know exactly how to exercise and hold the skill to do so while never initiating the behaviour; the deficit is motivational, not cognitive.

Theoretical accounts have moved through three broad generations, each correcting the last.

- **Drive and homeostasis.** Early accounts treated behaviour as the reduction of biological deficits, with homeostasis restoring set points. This handles hunger, thirst, and temperature well (and connects directly to the biological bases treated elsewhere), but poorly explains behaviour that increases arousal or that occurs in the absence of any deficit.
- **Incentive and expectancy-value theories.** The emphasis shifted to the pull of anticipated outcomes. Behaviour is predicted by expectancy of success multiplied by the value attached to the outcome — a formulation that remains the backbone of modern models of achievement, health, and consumer choice.
- **Contemporary goal and self-regulation accounts.** These specify how abstract values are converted into action: goals must be specific, sufficiently difficult to demand effort, committed to, and accompanied by feedback.

Several distinctions within motivation recur across domains:

Intrinsic versus extrinsic motivation. Intrinsic motivation arises from the activity itself; extrinsic from separable consequences. Self-determination theory identifies three basic psychological needs — autonomy, competence, and relatedness — whose satisfaction supports intrinsic motivation and internalisation of extrinsic goals. The theory's most consequential and contested claim is that controlling

rewards can *undermine* intrinsic motivation, shifting the perceived locus of causality from within to outside.

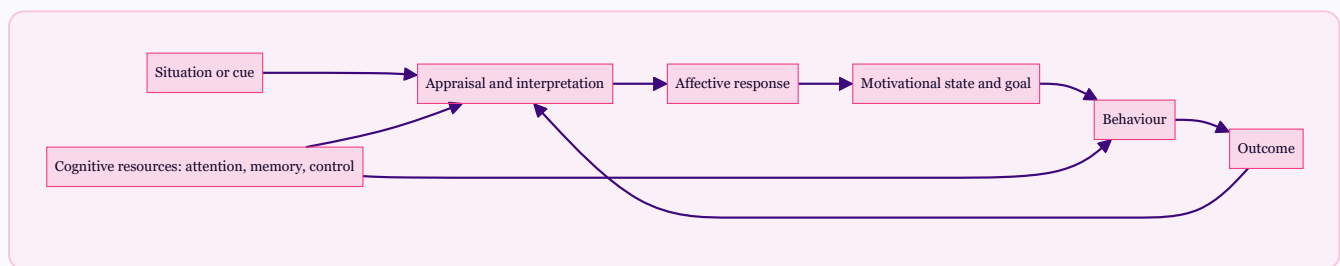
Wanting versus liking. Incentive-salience accounts separate the motivational pull of a reward from its hedonic impact. Cue-triggered "wanting" can escalate independently of "liking", which illuminates compulsive patterns in which a behaviour is pursued vigorously while yielding diminishing pleasure.

Self-efficacy. Bandura's construct — the person's belief in their capacity to execute the required action — predicts initiation, effort, and persistence, frequently more strongly than outcome expectancies. Crucially, it is domain-specific and modifiable through mastery experience, vicarious modelling, persuasion, and reinterpretation of arousal.

Time and discounting. Where outcomes are delayed, their motivational force decays, typically hyperbolically, producing preference reversals: a smaller-sooner option chosen now that would be rejected in advance. This is the mechanism treated in the chapter on Present Bias and Self-Control, and it explains why commitment devices outperform good intentions.

The Integrative View: Appraisal, Affect, Motivation, and Control

The three families are best modelled as a feedback loop in which cognition supplies the appraisal, affect supplies the value signal, motivation supplies the direction, and outcomes update the whole system.



The loop is not merely heuristic. It generates testable predictions: manipulating appraisal should change affect and, downstream, behaviour; manipulating affect should change the effort an organism is willing to expend; and manipulating outcomes should change subsequent appraisal, producing the self-reinforcing patterns familiar in clinical and organisational settings.

Worked example: a student who cannot start revising

A postgraduate student with an examination four weeks away repeatedly postpones revision, working instead on low-priority tasks. Analysed within the loop:

- **Cognitive.** The task is represented abstractly ("revise the whole syllabus"), with no specific next action, so attention drifts to stimuli offering immediate feedback. Working memory load from anxiety further reduces the resources available for planning. Prior failures are highly available and inflate the estimated difficulty.
- **Affective.** Primary appraisal registers the examination as a high-stakes threat; secondary appraisal judges coping resources inadequate. The resulting anxiety is aversive, and avoidance terminates it immediately — an affective relief that is itself reinforcing.
- **Motivational.** Expectancy of success is low while task value remains high, a combination that predicts disengagement rather than effort. Intrinsic motivation is negligible; the goal lacks specificity and

feedback; the salient comparison is between immediate discomfort and a distant, uncertain gain, which present bias weights heavily in favour of postponement.

The integrated account also identifies intervention points, each corresponding to a different level of the loop:

1. **Reappraise the task** — reframe the examination from a threat to a challenge by specifying what is controllable, lowering threat appraisal (cognitive change).
2. **Restructure the goal** — convert the abstract intention into specific, proximal, feedback-bearing sub-goals, restoring efficacy signals (motivation).
3. **Change the situation** — reduce the availability of immediately reinforcing alternatives and pre-commit to fixed sessions (situation selection, operating before the emotion is generated).
4. **Tolerate rather than escape the affect** — pair brief exposure to the aversive state with continued work, so that relief is no longer contingent on avoidance (affective regulation).

Note that no single item is the "real" cause, and no single intervention is guaranteed: the levels constrain one another, which is why multi-component interventions generally outperform single-strategy ones.

Measurement and Boundary Conditions

Because these constructs are not directly observed, their scientific standing depends on operationalisation. The following table illustrates typical conversions of construct into measurable index; the general problems of validity, reliability, and reactivity are treated in Research Methods and Applied Behavioural Science.

Construct	Typical operationalisation	Behavioural index
Working memory	Complex span tasks	Persistence on demanding tasks
Attentional bias	Dot-probe, eye tracking	Avoidance of threatening material
Affect	Self-report scales, facial EMG, autonomic measures	Approach/avoidance, help-seeking
Emotion regulation	Self-report strategy questionnaires, task-based reappraisal	Recovery after stressors
Self-efficacy	Domain-specific scales	Initiation, effort, persistence
Motivation strength	Choice, willingness to pay, effort discounting	Rate or duration of performance

Three boundary conditions deserve emphasis. **Individual differences** are substantial and stable enough to predict behaviour across settings. **Culture** shapes which appraisals are available and which affective expressions are licit, so constructs transported across populations require re-validation. **Context** is not noise but a determinant: the same person, with the same beliefs and dispositions, will act differently under different incentives, cues, and social settings — the interactionist commitment noted at the outset.

Summary

- Cognition, affect, and motivation are psychological-level constructs that mediate between situation and behaviour; they are admissible insofar as they are operationalised and measurably predictive.
- Cognition comprises attention, memory, schema-driven interpretation, executive control, and judgement; its behavioural signature is that identical stimuli yield different responses depending on what is known and expected.
- Affect spans emotion (brief, object-directed, multi-componential), mood (diffuse, prolonged), and stress (demand–resource mismatch). Appraisal theory locates the cognition–affect junction; affect in turn biases memory, judgement, and effort.
- Emotion regulation is a sequence of opportunities; earlier, antecedent-focused strategies generally carry lower costs than late suppression of an already-generated response.
- Motivation selects and sustains behaviour. Accounts have moved from drive reduction, through expectancy-value formulations, to goal and self-regulation models distinguishing intrinsic from extrinsic motivation, wanting from liking, and immediate from delayed outcomes.
- The three families form a feedback loop: appraisal generates affect, affect and expectancy shape motivation, motivation directs attention and effort, and outcomes revise the appraisal. Interventions may target any node.
- Measurement requires explicit operationalisation, and the generality of findings is bounded by individual differences, culture, and context.

Learning, Conditioning, and Behaviour Change

Learning is the mechanism by which experience is converted into durable changes in what an organism does. It is the bridging concept of behavioural science: the process that connects environment to behaviour and, in doing so, supplies the discipline with its most powerful technology of change. This subsection treats learning as a set of experimentally specified processes — non-associative, classical, operant, and observational — and then examines how those processes are deliberately harnessed in applied behaviour change.

Defining Learning

Learning is a relatively enduring change in the *potential* for behaviour that results from experience. Each clause of that definition does work:

- *Relatively enduring* excludes transient states. A rat is not "learning" when it slows after fatigue or a drug; the change must persist beyond the conditions that produced it.
- *Potential for behaviour* distinguishes learning from performance. An animal may have learned a route without currently running it — Tolman's classic demonstration that rats given unrewarded exploration of a maze later navigated it as efficiently as rats rewarded throughout. Learning is an intervening variable inferred from behaviour, not identical with it.
- *Resulting from experience* excludes maturation. The onset of walking in a toddler reflects neuromuscular development as much as practice; the two are dissociable and often confounded.

Two traditions of specification coexist. In the **associative** paradigm, learning is described as the formation of relations between events — between a cue and a reinforcer, or between a response and its consequence. In the **functionalist** tradition, learning is described as the calibration of behaviour to the local statistics of an environment, with whatever internal machinery achieves it. Both descriptions are used throughout this book, and Tinbergen's framework (see *Definition and Scope of Behavioural Science*) clarifies why they do not compete: they answer mechanistic and functional questions respectively.

Non-Associative Learning

Not all learning involves relations between events. **Habituation** is a progressive decline in a response to a repeated stimulus — the ticking clock that ceases to be noticed, the startle response that wanes over repeated tones. **Sensitisation** is the converse: an increase in responsiveness, typically following an intense or noxious event, such that subsequent stimuli elicit exaggerated reactions.

The dual-process theory of Groves and Thompson (1970) treats habituation and sensitisation as independent processes operating simultaneously in the stimulus–response pathway and in a general state system; the observed behaviour is their sum. This accounts for dishabituation (a novel or intense stimulus temporarily restores the habituated response) and for the fact that sensitisation fades quickly, whereas habituation is more durable. It also distinguishes habituation from sensory adaptation and from effector fatigue, neither of which depends on the organism's history in the way habituation does:

Phenomenon	Locus	Depends on history?	Recovers with rest?
Habituation	Central	Yes	Yes, spontaneously
Sensory adaptation	Receptor	No	Within seconds to minutes
Fatigue	Effector	No	With metabolic recovery
Sensitisation	Central state system	Yes	Rapidly

Non-associative processes matter practically because habituation is the mechanism by which benign stimuli — a new commute, a mild annoyance, a mild symptom — lose their behavioural grip, and sensitisation explains why stress amplifies irritability across domains rather than in one domain alone.

Classical Conditioning

Classical (Pavlovian) conditioning is the paradigm in which a neutral stimulus that reliably predicts a biologically significant event acquires the capacity to elicit a response. Pavlov's vocabulary is now standard: the **unconditioned stimulus** (US) elicits the unconditioned response (UR) without training; a **conditioned stimulus** (CS) is initially neutral; after pairing, the CS alone elicits the **conditioned response** (CR).

The CR is not simply a weaker UR. Pavlov's own observations, and later work on drug tolerance by Siegel (1975), show that conditioned responses are frequently *compensatory*: a situational cue paired with an opiate elicits a conditioned reaction that partially counteracts the drug's effect. This explains the pharmacological fact that tolerance is context specific, and the clinical fact that a former user is at greatest risk of overdose in a novel setting — the familiar cues that normally blunt the drug's action are absent.

Several determinants of conditioning were established experimentally and are widely used in applied settings:

- **Contiguity and contingency.** Rescorla's (1968) experiments showed that pairing alone is insufficient; the CS must *predict* the US more reliably than other cues do. Conditioning depends on contingency, not merely on temporal pairing.
- **Blocking and competition.** Kamin (1969) showed that a cue which is redundant with an already informative CS learns little; Pavlov's overshadowing shows that the more salient of two simultaneous cues dominates. Cues compete for predictive value.
- **Latent inhibition.** Pre-exposing a stimulus without consequence retards its later acquisition (Lubow & Moore, 1959). Familiarity can be a barrier to association.
- **Higher-order conditioning and conditioned inhibition.** Cues can acquire power through other conditioned cues, and can also acquire inhibitory properties, suppressing a response when they signal the absence of the US.

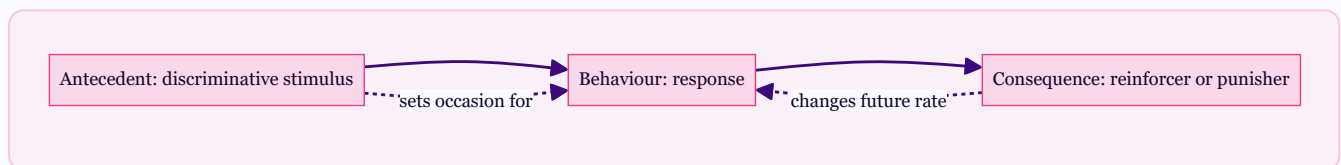
Extinction is new learning

Repeated presentation of the CS without the US produces **extinction** of the conditioned response. A persistent misconception — one shared by clinicians and clients alike — is that extinction erases the original association. It does not. The acquisition memory survives, and evidences itself in **spontaneous recovery** (the response returns after a delay), **renewal** (it returns when testing occurs in a different context), and

reinstatement (it returns after an unsignalled US). Contemporary exposure therapy therefore treats extinction as the construction of a new, context-sensitive inhibitory memory that must be retrieved to compete with the old one — a framework known as inhibitory learning (Craske and colleagues, 2014), with direct implications for how exposure is designed and varied.

Operant Conditioning

Operant (instrumental) conditioning concerns the relation between a response and its consequences. Thorndike's law of effect, derived from puzzle-box studies (1898), held that responses followed by satisfying consequences are strengthened. Skinner (1938) formalised the relation as the **three-term contingency**: a discriminative stimulus sets the occasion for a response, which produces a consequence.



This A–B–C unit is the analytical basis of applied behaviour analysis: behaviour is understood and changed by identifying the antecedents that occasion it and the consequences that maintain it.

Reinforcement and punishment

Two dimensions generate four operations: consequence type (delivered or removed) and effect (behaviour increases or decreases).

Operation	Consequence	Effect on behaviour	Example
Positive reinforcement	Stimulus added	Increases	Bonus paid for sales targets
Negative reinforcement	Stimulus removed	Increases	Buckling a seatbelt to silence an alarm
Positive punishment	Stimulus added	Decreases	Fine for illegal parking
Negative punishment	Stimulus removed	Decreases	Loss of phone privileges

Two clarifications are routinely required. First, **negative reinforcement is reinforcement**: it strengthens behaviour. Learners commonly conflate it with punishment because both can involve aversive events; the distinction lies entirely in the direction of the behavioural effect. Second, delivery is not sufficient — a "reinforcer" is defined functionally, by its effect on behaviour, not by its presumed pleasantness.

Reinforcers are often **conditioned**: neutral events acquire reinforcing power through association with primary reinforcers, a process that scales to **generalised reinforcers** such as money, tokens, or points that can be exchanged for a wide range of goods. The **Premack principle** states that a higher-probability activity can reinforce a lower-probability one (access to gaming contingent on homework completion); the more general **response deprivation** formulation (Timberlake & Allison, 1974) holds that any activity becomes reinforcing when constrained below its free-operant baseline.

Schedules of reinforcement

How consequences are distributed in time and across responses determines the characteristic pattern of behaviour — arguably the most practically useful body of findings in the field.

Schedule	Delivery	Typical pattern	Resistance to extinction
Fixed ratio	After n responses	High rate with post-reinforcement pauses	Moderate
Variable ratio	After n responses on average	High, steady rate	High
Fixed interval	First response after fixed time	"Scalloped" acceleration	Low–moderate
Variable interval	First response after variable time	Moderate, steady rate	High

Two implications follow. **Intermittent reinforcement produces behaviour that is markedly more resistant to extinction** than continuous reinforcement (the partial reinforcement extinction effect) — the reason gambling schedules are so durable, and the reason patterns maintained on variable reinforcement are hard to disrupt. Conversely, **ratio strain** occurs when the ratio requirement escalates faster than the organism's capacity to sustain it, producing breakdown of the response rather than persistence.

Shaping, chaining, and stimulus control

Complex behaviour is built by **shaping** — reinforcing successive approximations — a technique as applicable to teaching a child to write letters as to training a rat to press a lever. **Chaining** links component behaviours into sequences by establishing each as a discriminative stimulus for the next. Discriminative stimuli come to control behaviour through differential reinforcement, producing **generalisation gradients** (responding to stimuli similar to the training stimulus) and sharp **discrimination** after training on closely spaced alternatives.

Choice and the matching law

When two or more responses are concurrently available, Herrnstein's (1961) **matching law** states that the proportion of responses allocated to an alternative matches the proportion of reinforcers obtained from it. Departures from strict matching — under-matching, bias toward one alternative — reflect the contributions of other reinforcers and of the effort costs involved. When reinforcement is delayed, its effective value decays hyperbolically (Mazur, 1987), which is the behavioural substrate of present bias treated in the chapter on *Present Bias and Self-Control*.

Biological Constraints on Conditioning

A purely general-process account of learning — in which any cue can be associated with any consequence given adequate contingency — is empirically false. Garcia and Koelling (1966) demonstrated that rats readily learn to associate a *taste* with induced illness but not a light-and-tone compound, whereas they readily associate the compound with shock. Seligman (1970) termed this **preparedness**; Bolles (1970) argued that animals bring **species-specific defence reactions** to aversive contingencies; and the

Brelands (1961) documented "instinctive drift", in which trained raccoons began "washing" their coins and pigs began rooting them, despite sustained reinforcement to the contrary.

The lesson is not that conditioning laws fail but that they are constrained by evolutionary history: what an animal learns readily is what its lineage was selected to learn. This convergence between learning theory and ethology is one reason the modern field treats associative and evolutionary accounts as complementary rather than rival.

Cognitive Elaborations of Conditioning

Learning theory absorbed cognitive constructs rather than being displaced by them.

- **Expectancy and cognitive maps.** Tolman's latent-learning and place-learning studies showed that animals acquire spatial knowledge that is not reducible to chains of stimulus–response links.
- **Prediction error.** The Rescorla–Wagner (1972) model formalised conditioning as the reduction of a discrepancy between the reinforcer received and that expected. Schultz's (1997) recordings of midbrain dopamine neurons showed that their firing tracks precisely this prediction error, giving associative learning a neural implementation — a striking demonstration of multilevel convergence (see the levels of analysis framework in the parent chapter).
- **Learned helplessness.** Seligman and Maier (1967) found that dogs exposed to inescapable shock later failed to escape avoidable shock. Maier and Seligman's later reformulation (2016) holds that passivity is the default response to prolonged aversive events and that what is actually *learned* is the detection of control; uncontrollability is the pathological condition. The construct has been applied — sometimes loosely — to depression, work disengagement, and institutional passivity.
- **Rule-governed behaviour.** Human performance is often controlled by verbal descriptions of contingencies rather than by direct exposure to them, which makes it efficient, transferable, and sometimes *insensitive* to changed contingencies — a persistent theme in applied settings where instructions and consequences diverge.

Observational and Social Learning

Bandura's Bobo doll experiments (1961) showed that children reproduce aggression they have merely observed, without direct reinforcement. Bandura's social learning theory (1977) identified four subprocesses: **attention**, **retention**, **reproduction**, and **motivation** (itself affected by vicarious reinforcement, direct reinforcement, and self-evaluation). The construct of **self-efficacy** — belief in one's capacity to execute the behaviour — became one of the most widely applied concepts in behaviour change.

Comparative work distinguishes **imitation** (copying the specific means) from **emulation** (achieving the same outcome by any means) and from **stimulus enhancement** (attention being drawn to an object). Claims about mirror neurons as a dedicated imitation system are best treated cautiously; the evidence supports a role in action observation and prediction rather than a single mechanism for cultural transmission.

From Principles to Behaviour Change

Applied behaviour change is the deliberate arrangement of antecedents and consequences — and, increasingly, of the cognitive and social variables above — to alter a target behaviour. A disciplined sequence is used across health, education, clinical, and policy contexts.

1. **Specify the behaviour.** Define the target operationally, with a measurable unit (episodes per day, minutes, proportion of opportunities taken). Vague targets ("be healthier") are not changeable.
2. **Conduct a functional analysis.** Identify antecedents, behaviour, and maintaining consequences. This distinguishes, for instance, behaviour maintained by attention from behaviour maintained by escape, which require different interventions.
3. **Select techniques.** The Behaviour Change Technique Taxonomy v1 (Michie and colleagues, 2013) catalogues 93 discrete, describable techniques — goal setting, self-monitoring, prompts and cues, feedback, graded tasks, restructuring the physical environment. The COM-B model and Behaviour Change Wheel (Michie et al., 2011) map interventions to whether the binding constraint is capability, opportunity, or motivation.
4. **Implement and measure.** Single-case designs — withdrawal (ABAB), multiple baseline, alternating treatments — permit causal inference with individual cases, which is often the only feasible unit of analysis.
5. **Plan for generalisation and maintenance.** Behaviour change that does not transfer across settings, or that decays on withdrawal of the intervention, has limited value. Relapse-prevention work attends explicitly to high-risk contexts, to the abstinence-violation effect, and to rebuilding after lapses.

Two families of technique

Antecedent-focused techniques alter the conditions under which behaviour occurs: changing the default, relocating a cue, prompting, implementation intentions (Gollwitzer, 1999, specifies *when, where, and how* a goal will be enacted, with medium-to-large effects in meta-analysis), and choice architecture. Their advantage is that they require no in-the-moment effort. **Consequence-focused** techniques alter what follows behaviour: contingency management, token economies, differential reinforcement, feedback, and incentive schemes.

Habit formation is the point of convergence. With repetition in a stable context, behaviour becomes cue-driven and correspondingly insensitive to variations in motivation; Lally and colleagues (2010) found a median of 66 days for a health behaviour to approach automaticity, with wide individual variation. Habits are powerful maintenance devices, and equally resistant to change once established — a symmetry that explains both the success of well-designed defaults and the difficulty of breaking entrenched routines.

Worked example: reducing late-night phone use in a graduate student

Target behaviour: episodes of phone unlocking after 22:30, logged automatically; baseline 41 per night.

Functional analysis: the phone is on the desk within reach (antecedent); the behaviour yields immediate relief from work-related anxiety and immediate social reward (consequence); the relevant state is fatigue and low self-regulatory capacity.

Technique selection: - *Antecedent restructuring:* charger moved to the hallway; notifications disabled after 21:30; an analogue alarm clock substituted. - *Implementation intention:* "If it is 22:15 and I am at my desk, then I will put the phone on the hallway charger and open a paper book." - *Consequence:* a shared log with a peer, with a small forfeit for nights exceeding five unlocks (positive punishment) and a weekend reward for a streak (positive reinforcement on a fixed-ratio-of-days schedule). - *Substitution:* a competing response (reading) with its own reinforcement history.

Measurement: ABAB withdrawal design across four weeks. Unlocks fall to 6 per night in phase B, rise to 27 when the intervention is withdrawn, and fall again on reinstatement.

Maintenance: the antecedent change carries most of the effect because it removes the cue rather than requiring resistance to it. The contingency is faded to an intermittent schedule to increase resistance to extinction, and a lapse plan is specified in advance to forestall the abstinence-violation effect.

The example is instructive because no single ingredient is doing all the work: the environment does most of it, the contingency supplies initial momentum, and the plan protects against relapse.

Applied domains and their evidence

- **Clinical.** Exposure-based procedures for anxiety disorders are extinction protocols in all but name; contingency management with voucher incentives has among the strongest evidence bases in the treatment of stimulant use disorders; applied behaviour analysis is the best-validated intervention for skill acquisition in autism, though it has attracted sustained ethical criticism where it has targeted the suppression of harmless, identity-related behaviours or used aversive procedures.
- **Health.** Smoking cessation, medication adherence, and physical activity interventions rely heavily on self-monitoring, goal setting, and environmental restructuring; effect sizes are modest but population reach is large.
- **Education and workplace.** Token systems, feedback, and reinforcement of engagement; the pitfalls of extrinsically rewarding intrinsically motivated activity are well documented but conditional on how the contingency is framed.
- **Policy.** Defaults, reminders, and incentives — the ecological side of choice architecture discussed in *Behavioural Economics in Policy and Markets*.

Cautions

Three cautions are worth stating explicitly. First, **punishment is a weak technology**: it suppresses rather than replaces, its effects are context-bound and often rebound after removal, and it generates avoidance, aggression, and escape from the punishing agent. Consequence-based design is therefore usually reinforcement-based wherever feasible. Second, **behaviour change raises ethical questions that the science cannot answer**. Interventions that operate without the person's awareness may be effective and still objectionable on grounds of autonomy and transparency; the field now treats these questions as internal to its subject matter. Third, **effect sizes are typically small and heterogeneous**, and many published interventions fail to survive replication or to generalise beyond the original setting.

Summary

- Learning is a relatively enduring, experience-dependent change in behavioural potential; it is to be distinguished from maturation, transient states, and current performance.
- Non-associative processes — habituation and sensitisation — modify responsiveness without forming associations and are best understood as parallel, competing influences.
- Classical conditioning establishes predictive relations between stimuli. Contingency, not mere contiguity, drives acquisition; extinction creates a new inhibitory memory rather than erasing the original, which is why spontaneous recovery and renewal occur.
- Operant conditioning governs the relation between responses and consequences, formalised in the three-term contingency. Reinforcement strengthens; punishment suppresses; negative reinforcement is

reinforcement.

- Schedules of reinforcement determine the pattern and persistence of behaviour; intermittent schedules produce especially durable responding.
- Learning is biologically constrained: preparedness, species-specific defence reactions, and instinctive drift show that what is learned readily depends on evolutionary history.
- Cognitive constructs — expectancies, cognitive maps, prediction error, learned helplessness — were absorbed into learning theory rather than displacing it, and prediction error now links associative models to dopaminergic signalling.
- Observational learning broadens the account to socially transmitted behaviour, with self-efficacy a central mediator.
- Applied behaviour change proceeds by specifying behaviour, conducting a functional analysis, selecting techniques from a structured taxonomy, measuring with single-case designs, and planning for generalisation and maintenance.
- The most reliable interventions are usually antecedent-based and reinforcement-based; punishment is a weak and ethically fraught instrument; and ethical appraisal of behaviour-change interventions is part of the science, not an external constraint upon it.

Social, Cultural, and Environmental Influences

Behaviour is never emitted into a vacuum. Every act — a purchase, a vote, a greeting, a refusal — occurs in a setting populated by other people, governed by shared expectations, and structured by physical and informational environments that were often built by someone else. This subsection examines the three broad classes of external influence that shape behaviour: **social** (the actual, imagined, or implied presence of others), **cultural** (transmitted norms, meanings, and practices), and **environmental** (the physical, built, and digital settings in which behaviour unfolds). The unifying claim is that these are not mere background conditions but causal variables, and that they can be operationalised, measured, and manipulated with the same discipline applied to any other determinant of behaviour.

Social Influences: From Mere Presence to Shared Norms

The effects of mere presence

The oldest experimental question in social psychology is what happens when others are simply *there*. Zajonc's (1965) drive-theory resolution of a long-standing contradiction — that presence sometimes improves and sometimes degrades performance — proposed that the presence of others raises arousal, and that arousal strengthens the organism's dominant response. On well-learned tasks the dominant response is correct, producing **social facilitation**; on novel or difficult tasks it is incorrect, producing **social inhibition**. The account is a useful reminder that social variables typically operate through mechanisms already described elsewhere in this book: arousal, attention, and reinforcement history.

Presence effects are also automatic. Chartrand and Bargh (1999) demonstrated that people unconsciously mimic the postures and mannerisms of interaction partners, and that such mimicry increases liking and rapport. Emotional contagion — the spread of affect through faces, voices, and postures — supplies a fast channel by which one person's state becomes another's. Behavioural contagion extends further: in observational network analyses, health behaviours such as smoking cessation, obesity, and exercise have been shown to cluster in social ties, though the strength of these effects has been contested on the grounds that homophily and shared environment can mimic transmission. Distinguishing *influence* from *selection* is one of the field's persistent inferential problems, and it is the reason network studies increasingly rely on longitudinal and natural-experimental designs.

Modelling and vicarious learning

People acquire behaviour by observing its consequences for others. Bandura's modelling experiments established that children reproduce aggression they have merely witnessed, and that whether they reproduce it depends on the observed outcome — vicarious reinforcement and punishment. This is treated in detail in the chapter on learning and conditioning; for present purposes the important point is that the *social* environment supplies most of the contingencies of daily life, and it does so symbolically, without the observer paying the cost of direct experience.

Conformity, compliance, and obedience

Three classical programmes mapped the terrain of yielding to others.

- **Conformity.** Asch's (1951) line-judgement studies showed that a substantial minority of participants would endorse a perceptually false judgement to match a unanimous majority. Sherif's earlier

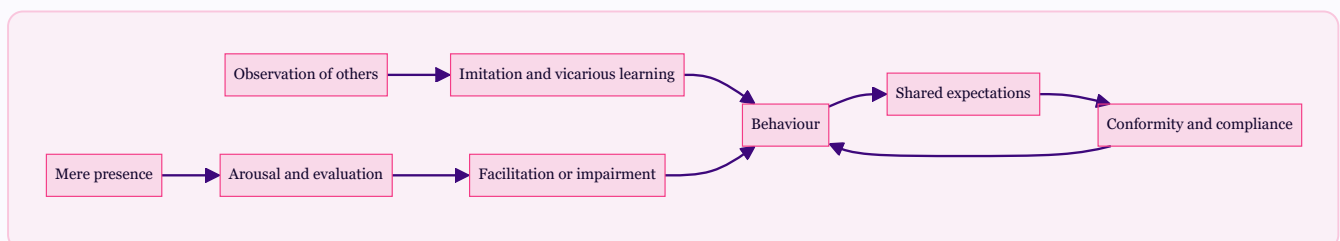
autokinetic work showed that judgements converge on a group norm that persists after the group disbands.

- **Compliance.** Research on sequential-request techniques — foot-in-the-door, door-in-the-face, and the exploitation of reciprocity — shows that the *form* of a request, not only its content, governs assent.
- **Obedience.** Milgram's (1963) experiments on instructed harm demonstrated that situational pressure from a legitimate-seeming authority can override personal standards at rates participants themselves did not predict.

These findings are commonly misread as evidence that people are simply weak. A more defensible reading, and the one now standard, is that the experiments identify *situational features* — unanimity, graduated commitment, an authority who accepts responsibility — that any adequate theory of behaviour must incorporate. Deutsch and Gerard's (1955) distinction remains the most economical organising device: **informational influence** occurs when others' responses are treated as evidence about reality, and **normative influence** occurs when they are treated as a source of social reward and sanction. The two predict different patterns of private acceptance and public compliance.

Group and collective processes

At aggregate levels, additional regularities appear. Social loafing shows that individuals exert less effort when contributions are pooled and unidentifiable. Deindividuation research links anonymity and reduced self-awareness to increased adherence to whatever local norm prevails — which may be prosocial or antisocial. Group polarisation shows that discussion tends to intensify the pre-existing leanings of a like-minded group rather than moderate them. The bystander literature, inaugurated by Darley and Latané (1968), established that the probability of intervention falls as the number of potential helpers rises, through diffusion of responsibility and pluralistic ignorance. Each of these is a small number of mechanisms producing large, counterintuitive, and practically consequential shifts in behaviour.



Norms: the connective tissue

The concept that links the social and cultural levels is the **social norm**: a shared expectation about conduct, held by a group, that is at least partly sustained by the anticipation of approval or disapproval. Cialdini's focus theory distinguishes **descriptive norms** (what most people do) from **injunctive norms** (what most people approve of), and by extension **injunctive norms**, and shows that a norm influences behaviour only when it is *salient* in the immediate situation. This is a strong and testable claim: interventions that make one norm focal can increase or decrease the same behaviour without changing any material incentive.

Norm misperception is pervasive. **Pluralistic ignorance** arises when individuals privately reject a norm but infer from others' public behaviour that it is widely endorsed; the resulting conservatism can freeze behaviour in place long after the underlying attitudes have shifted. Interventions that correct misperceived norms — telling students that their peers drink less than they assume, or that most employees do report

misconduct — therefore operate by changing the informational basis of behaviour rather than by adding incentives.

Cultural Influences

Culture as transmitted information

Culture may be defined, for behavioural purposes, as the set of norms, practices, symbols, and artefacts transmitted across individuals and generations, which shape what behaviour is expected, permitted, and intelligible. This definition is deliberately behavioural: culture is not a mystery variable invoked when other explanations fail, but a population-level distribution of information that reaches each individual through identifiable channels — caregivers, peers, institutions, media, and the built environment itself.

Two consequences follow. First, culture is learned and therefore variable; the same underlying person placed in different cultural contexts may behave differently. Second, culture is transmitted with bias. Research on cultural evolution identifies content and context biases that make some variants spread more readily than others: **conformity bias** (copy the majority), **prestige bias** (copy the successful), and **similarity bias** (copy those like oneself). These biases are mechanisms, not metaphors, and they generate testable predictions about the diffusion of practices.

Dimensions of cultural variation

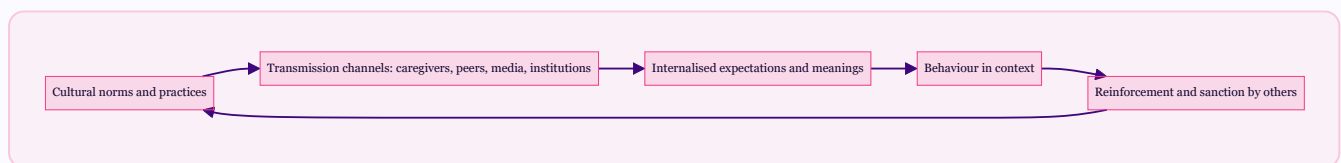
Cross-cultural psychology has sought to describe variation along a small number of dimensions. The most influential include individualism–collectivism (the relative weighting of personal goals versus ingroup obligations), power distance (the acceptance of hierarchical asymmetry), tightness–looseness (the strength of norms and tolerance of deviance; Gelfand and colleagues, 2011), and long-term versus short-term orientation. These dimensions are *aggregate* descriptors: they characterise distributions of behaviour within populations, and they license no inference about a particular individual, a caution that follows directly from the levels-of-analysis principle set out earlier in this chapter.

Cultural variation appears in processes that look universal at first glance. Attribution shows systematic differences: the tendency to overweight dispositional and underweight situational causes is weaker, or differently patterned, in many East Asian samples. Emotion display is governed by culturally transmitted **display rules** that regulate which expressions are appropriate in which settings. Norms of honour, reciprocity, and hospitality vary in force and in the sanctions that sustain them. Even the boundary between behaviour treated as a matter of personal choice and behaviour treated as a matter of collective obligation differs across contexts.

Dimension	Contrast	Illustrative behavioural consequence
Individualism–collectivism	Personal goals vs. ingroup obligations	Willingness to voice dissent; choice of consumption
Tightness–looseness	Strong vs. weak norms, low vs. high deviance tolerance	Rule-following under ambiguity; creativity
Power distance	Acceptance of hierarchy	Deference to authority; disclosure upwards
Display rules	Expressive regulation	Emotional expression at work and in public

The generalisability caveat

Much of the evidence base for the social and cultural claims above derives from participants in Western, educated, industrialised, rich, and democratic societies — the so-called **WEIRD** populations that are atypical on several psychological dimensions relative to the global majority. This is a constraint on external validity, not a reason to discard findings, but it does mean that generalisation must be treated as an empirical question rather than an assumption. Cultural priming studies (for example, work with bicultural participants who shift between cultural frames depending on which cues are salient) further suggest that cultural influence is often *situated* rather than dispositional: the relevant norms are activated by context.



Environmental and Situational Influences

The situation as a causal variable

Lewin's formulation that behaviour is a function of the person and the environment together (1936) remains the working assumption of the field: behaviour is jointly determined by the attributes an individual brings to a setting and the features the setting affords. The historical swing that made this a live issue was the **person–situation debate**, in which Mischel (1968) argued that cross-situational consistency in behaviour is far lower than trait models imply. The debate was resolved not by awarding victory to either side but by adopting **interactionism**: traits predict behaviour *differentially across situations*, and situations differ in how much they constrain behaviour at all.

Two moderators make this precise. **Situational strength** refers to the clarity and force of cues that indicate what behaviour is expected and rewarded; strong situations — formal examinations, military drill, queueing — suppress individual differences, while weak situations allow dispositions to express themselves. **Trait activation** specifies that a disposition is expressed only when the situation provides cues relevant to it: an extraverted person is expressive at a party but not at a silent retreat. Both concepts are directly relevant to intervention design, since a behaviour that is under strong situational control will not be moved by appeals to character.

Physical and ambient conditions

A substantial literature documents effects of the ambient environment on behaviour:

- **Noise.** Unpredictable and uncontrollable noise degrades performance on complex tasks, reduces helping, and impairs post-task persistence; the pattern follows from attentional overload and learned helplessness rather than from loudness alone.
- **Crowding and density.** Perceived crowding, as distinct from objective density, elevates stress indices and withdrawal, and the effect is moderated by perceived control and by social norms governing personal space.
- **Thermal conditions.** The heat–aggression relation is among the more replicated ambient effects, most clearly for irritability-based aggression, and is one of the rare areas where climate change is studied as a behavioural variable.
- **Light and nature.** Daytime light exposure and views of greenery are associated with improved mood, attention, and recovery. Attention restoration theory and Ulrich's hospital-window studies provide the standard accounts.
- **Air quality.** Poor ventilation and elevated particulate levels are associated with measurable decrements in cognitive performance, an effect of growing policy interest.

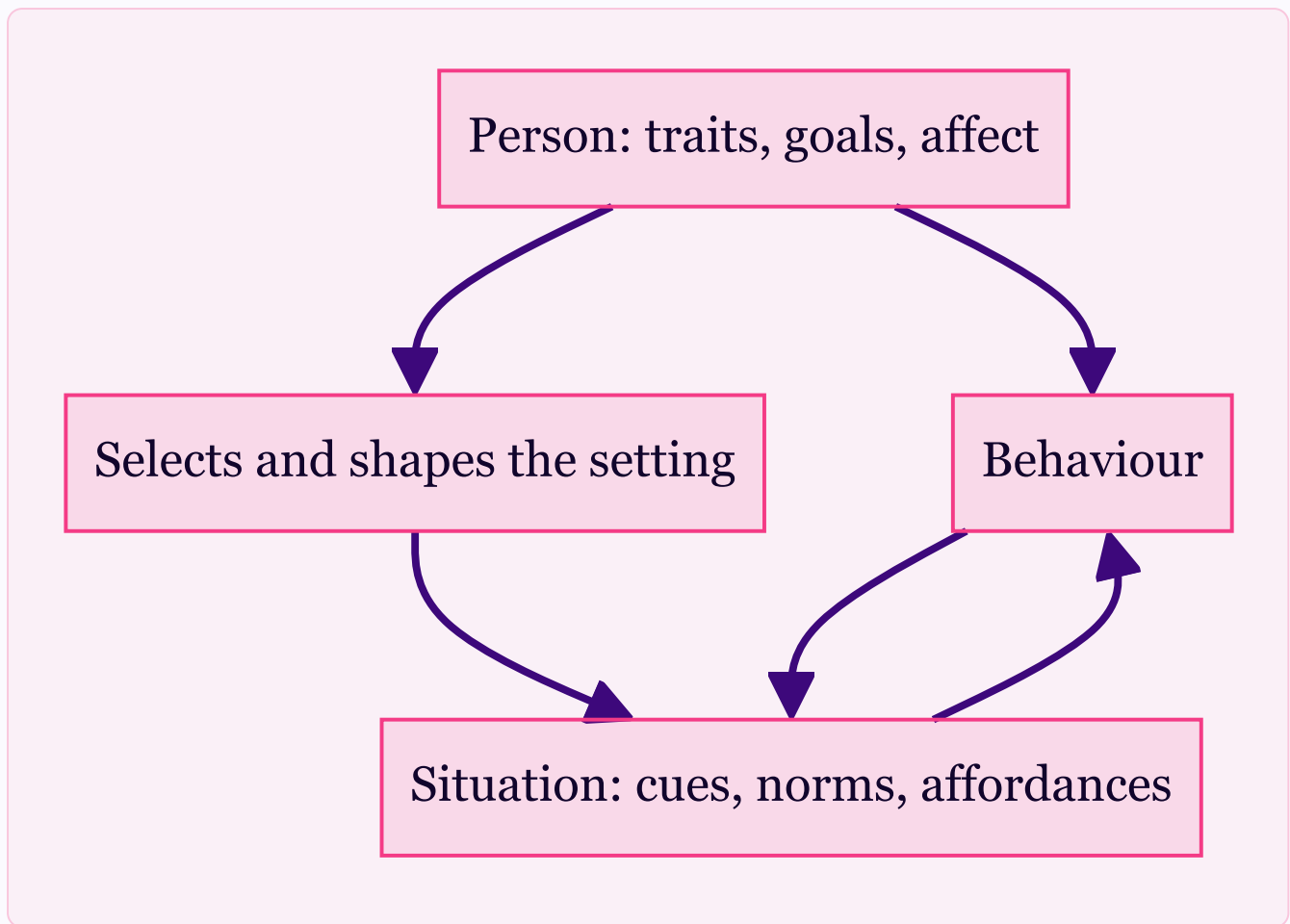
The built environment and choice architecture

Environmental influence is frequently *designed*. The placement of goods, the width of a corridor, the presence of a bin within reach, the visibility of a staircase, and the default setting on a form are all environmental features that alter behaviour rates without altering preferences or prices. Four overlaps with neighbouring material are worth flagging: these design features are the stock-in-trade of the chapters on behavioural economics and applied extensions; the layout of a setting that constrains behaviour is what ecological psychology describes in terms of **affordances** and behaviour settings; and the notion of a **default** is treated formally in the behavioural economics material.

Caution is warranted on the strength of such effects. Wilson and Kelling's (1982) *broken windows* thesis — that visible disorder invites further disorder — became enormously influential in policing, but subsequent field-experimental and reanalysis work has found the evidence weaker and more conditional than the original argument implied: disorder cues matter in some settings and for some outcomes, and the causal chain to serious crime is contested. The methodological lesson is general: environmental claims based on cross-sectional correlation between area characteristics and behaviour rates are vulnerable to confounding by selection and by unmeasured differences in enforcement and residents.

Digital environments

A large and growing share of behaviour now occurs in environments engineered to shape it. Digital settings supply: **cues** that make norms salient (visible like counts, ratings, and follower statistics); **friction** or its absence (one-click purchase, pre-checked consent, autoplay); **social comparison** against curated reference groups; and **algorithmic selection** that determines what information reaches the person. These are environmental variables in the ordinary sense of the term, and they invite the same questions of description, prediction, causal explanation, and intervention that the field applies elsewhere — alongside the ethics of transparency and consent discussed later in this subsection.



Studied How? Evidence and Inference

The influences described here are difficult to study precisely because they are not randomised by nature: people choose their friends, neighbourhoods, media, and workplaces. The field's responses are familiar from the chapters on research methods — field and natural experiments, randomisation at the level of the setting rather than the person (classrooms, floors of a building, postal codes), ecological momentary assessment to capture behaviour in situ, and multilevel models that separate individual from contextual variance. Cross-cultural work adds requirements of measurement equivalence, since a scale that behaves differently in two populations cannot be used to compare them. Where randomisation is impossible, the strategy is to triangulate across designs with different bias structures rather than to place weight on any single correlation.

Worked example: social norms and household energy use

Schultz and colleagues (2007) provided households with information about their own energy consumption and that of their neighbours. The descriptive-norm message worked as a *pull toward the mean*: below-average consumers **increased** their consumption. The addition of an injunctive element — a simple approval cue such as a smiley face — eliminated this boomerang effect while preserving the reduction among high consumers. The example is instructive on three counts. It demonstrates that a social norm is a causal variable with a measurable effect size; it demonstrates that the *framing* of a norm message determines the sign of the effect; and it shows why social-level interventions require the same careful evaluation as individual-level ones, since an intuitively appealing message produced a perverse outcome on its own.

Worked example: bystander intervention by design

The bystander effect becomes an actionable environmental problem once it is described mechanically. Intervention strategies follow from the mechanism rather than from exhortation: reduce the ambiguity of the situation by making the need for help explicit and unmistakable; reduce diffusion by ensuring identifiability of responsibility; and supply a decision rule that requires no deliberation. Practical implementations include the "if you see something, say something" campaigns in transit systems, designated bystander roles in workplace harassment training, and deliberate single-responder protocols in emergency call centres. Each modifies the situation rather than the persons in it.

Boundaries, Ethics, and Normative Limits

Intervening on behaviour through social norms, cultural framing, or environmental design raises issues that the field treats as internal to its subject matter. Three are recurrent. **Autonomy and transparency:** influence that operates without the target's awareness — a default, a norm cue, a redesigned corridor — differs from persuasion in ways that bear on consent, and the distinction between "nudge" and "manipulation" is contested rather than settled. **Cultural fit:** an intervention validated in one population may be ineffective, or actively harmful, in another, particularly where it presumes the salience of individual choice in a setting where collective obligation governs. **Distributional effects:** because norms and environments are unequally distributed, interventions can widen rather than narrow existing behavioural gaps even when average outcomes improve.

It bears repeating, in light of the boundary drawn earlier in this chapter, that identifying these mechanisms does not settle whether the behaviours produced are desirable. Behavioural science can establish that a norm message reduces littering; whether littering ought to be reduced, by whom, and at whose cost, are questions it informs but cannot answer.

Summary

- Social, cultural, and environmental variables are causal determinants of behaviour, operationalisable and manipulative like any others, not mere background noise.
- Social influence ranges from automatic effects of mere presence and mimicry, through modelling and vicarious learning, to conformity, compliance, obedience, and group-level phenomena such as loafing, polarisation, and diffusion of responsibility.
- Social norms — descriptive and injunctive — regulate behaviour chiefly when salient, and misperceived norms sustain behaviours that individuals privately disapprove of, a phenomenon known as pluralistic ignorance.
- Culture is best treated as transmitted information: norms, practices, and meanings propagated through identifiable channels with identifiable transmission biases.
- Cultural variation is real but aggregate; individual inference from group-level dimensions is invalid, and the WEIRD composition of the evidence base constrains generalisation.
- Ambient conditions (noise, crowding, heat, light, air quality) and designed environments (placement, defaults, affordances, digital interfaces) exert reliable effects, though environmental claims resting on cross-sectional correlations are vulnerable to selection and confounding.
- Person and situation interact: situational strength and trait activation specify when dispositions predict behaviour and when environmental redesign is the more effective lever.

- Applications that shape behaviour socially, culturally, or environmentally raise unresolved questions of autonomy, cultural fit, and distributional fairness, which the field now treats as part of its subject matter.



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Research Methods and Applied Behavioural Science

Research methods are not an appendage to behavioural science; they are the mechanism by which the field earns the designation *science* in the first place. If behaviour is the dependent variable — observable, measurable, temporally extended activity, as fixed in the earlier discussion of the field's scope — then the methods described here are the procedures by which that activity is rendered countable, comparable, and causally interpretable. This subsection treats three connected matters: how behavioural constructs are operationalised and measured, how designs are chosen to license different strengths of inference, and how findings travel from the laboratory or field site into applied practice. The distinctive methodological difficulties flagged in that earlier discussion — the reactive subject, the impossibility of randomising life histories, the multiplicity of causes — recur throughout and structure the solutions on offer.

Operationalisation: From Construct to Indicator

A behavioural construct that cannot be tied to an indicator cannot be tested. Operationalisation is the translation of a theoretical term ("impulsivity," "adherence," "trust") into one or more measurable events, and the choice of indicator is consequential: different operationalisations of the same construct frequently produce only moderately correlated measures and sometimes yield different substantive conclusions.

Behavioural measures can be classified by their *response dimension*, a scheme inherited from the experimental analysis of behaviour and now standard across the field:

Dimension	What is recorded	Example
Frequency or rate	Count per unit time	Cigarettes smoked per day
Latency	Time to onset	Delay before responding to a warning
Duration	Time engaged	Minutes spent on a task
Intensity or amplitude	Force, effort, magnitude	Grip force on a hand dynamometer
Choice or allocation	Selection among options	Proportion of tokens invested
Topography	Physical form of the act	Sequence of keystrokes in a typing task

These dimensions may be recorded through direct observation with structured coding schemes, through automated instrumentation (actigraphy, eye tracking, infrared counters), through performance tasks with reaction-time or accuracy outcomes, through self-report inventories and ecological momentary assessment, through physiological indices (skin conductance, pupillometry, neuroimaging), or increasingly through *digital trace data* — clickstreams, transaction logs, geolocation, and platform telemetry. Each carries a characteristic bias. Self-report is vulnerable to recall error, social desirability, and demand characteristics; observational coding is vulnerable to observer drift and expectancy; physiological measures are sensitive but often ambiguous as to construct; digital traces are unobtrusive and vast but frequently collected for commercial rather than scientific purposes, which raises both measurement and ethical problems.

Two psychometric properties govern whether a measure is usable. **Reliability** concerns consistency — inter-rater agreement (conventionally indexed by Cohen's kappa or intraclass correlation), test–retest

stability, and internal consistency. **Validity** concerns whether the measure means what it is claimed to mean; the relevant subtypes include content, convergent, discriminant, predictive, and criterion validity. Reliability is necessary but not sufficient for validity: a scale can be internally consistent and consistently wrong.

A distinctive problem for behavioural measurement is **reactivity** — the capacity of the measurement procedure itself to alter the behaviour measured. An observer changes what is observed; a step counter changes walking; consenting to a study changes consumption. Standard mitigations include unobtrusive or archival measurement, habituation periods before data collection, blind coding, and the use of cover tasks, each of which must be weighed against the ethical requirements of informed consent and debriefing discussed below.

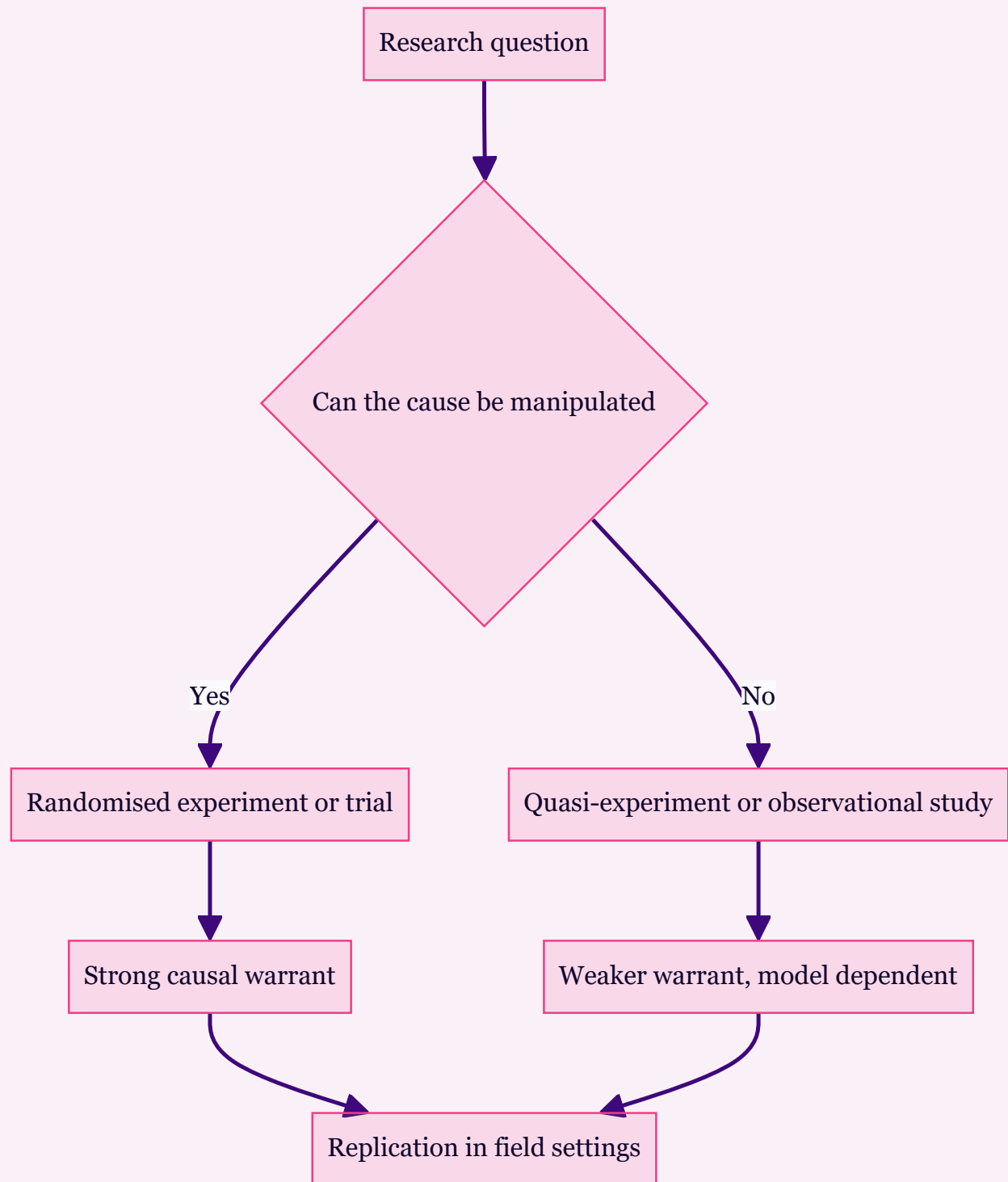
Observational and Descriptive Designs

The threefold ambition of description, prediction, and causal explanation described earlier imposes an ordering on designs. Descriptive work comes first, and it is more demanding than it appears. Naturalistic observation requires a sampling rule — focal-animal versus scan sampling, continuous recording versus interval or time sampling, event-contingent versus fixed-interval schedules — and the rule determines what can later be claimed. Ethograms and structured coding manuals make observations comparable across observers and sites; inter-observer agreement should be reported rather than assumed.

Descriptive designs are also the basis of behavioural epidemiology: prevalence and incidence of behaviours, their distribution across populations and time, and their covariation with demographic and environmental variables. Such work is often dismissed as merely descriptive, but it disciplines prediction and supplies the base rates without which causal claims cannot be evaluated.

Correlational, Quasi-Experimental, and Experimental Designs

The central inferential question is always the same: how much causal warrant does the design support? Randomisation is the strongest available warrant, because it renders treatment assignment independent of all covariates, measured and unmeasured. But life histories, institutions, and many hazards cannot be randomised, so the field relies on a graded family of designs.



Design	Assignment	Principal use	Main threat to inference
Naturalistic observation	None	Description, base rates	Observer bias, sampling artefact
Cross-sectional survey	None	Prevalence, association	Confounding, reverse causation
Longitudinal panel or cohort	None	Temporal ordering, development	Attrition, period effects
Twin and adoption designs	Natural	Heritability, gene–environment interplay	Assumption of equal environments
Interrupted time series, difference-in-differences, regression discontinuity	Natural or administrative	Policy and programme evaluation	Secular trends, assumption violations
Between-subjects randomised experiment	Random	Causal mechanism, efficacy	Demand characteristics, low power
Within-subjects experiment	Random order	Sensitive comparisons	Order, carryover, practice effects
Cluster or stepped-wedge trial	Random at group level	Field and service evaluation	Contamination, low effective sample size
Single-case experimental design	Reversal or staggered onset	Individual-level change, clinical practice	Order effects, limited generalisation

The threats to internal validity catalogued by Campbell and his collaborators remain the working checklist: history (events co-occurring with the intervention), maturation, testing, instrumentation, regression to the mean, selection, and differential attrition. Each has a design countermeasure — control series, waiting-list comparison, stable instruments, multiple pre-measurements, matching or statistical adjustment, and intention-to-treat analysis respectively.

Experiments add a further layer of control. Between-subjects designs avoid carryover but are less sensitive; within-subjects designs gain power but require counterbalancing to neutralise order effects. Factorial designs allow the detection of interactions, which are frequently the substantive finding of interest: an intervention that works for one population or setting and backfires in another is an interaction, not a failed replication. Blinding of participants, of those delivering the manipulation, and of analysts addresses expectancy effects, though true blinding is often impossible in behavioural interventions, where the participant necessarily knows whether they received the programme.

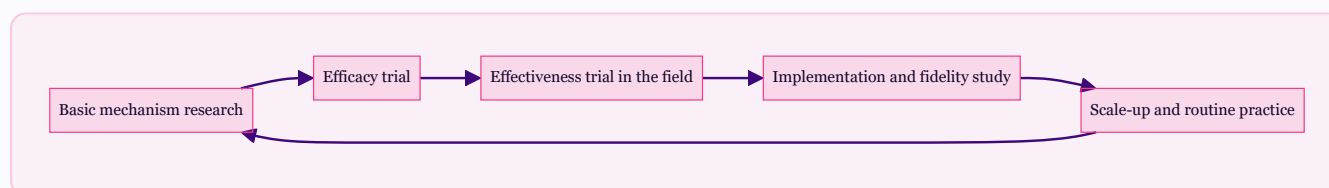
Single-case experimental designs deserve separate mention because they are the field's native method for individual-level causal inference and remain central to applied behaviour analysis and clinical practice. The reversal (ABAB) design alternates baseline and intervention phases to demonstrate that behaviour tracks the contingency; multiple-baseline designs stagger the introduction of the intervention across participants, settings, or behaviours, so that change coincides with onset in each series. These designs

substitute replication across cases and phases for replication across randomly assigned groups, and they are particularly valuable where withholding treatment is unethical or where cases are rare.

Sampling, Power, and Generalisation

Every design is only as good as the sample it runs on. Probability sampling (simple, stratified, cluster) supports inference to a defined population; convenience, snowball, and panel sampling — the workhorses of academic behavioural research — do not, and the resulting constraint on generalisability is the "WEIRD sample" problem noted among the field's open questions. Field experiments typically accept opportunistic samples in exchange for ecological validity and scale.

Two further issues deserve emphasis. First, **statistical power**: behavioural effects are usually small, and underpowered studies produce unstable estimates and inflated effects, a bias aggravated by selective reporting. Second, **multiplicity**: flexible analysis of many outcomes and subgroups manufactures findings. The methodological reforms prompted by the replication crisis — pre-registration and registered reports, open data and materials, pre-specified primary outcomes, multi-site replication, and explicit reporting of effect sizes with confidence intervals rather than p -values alone — are best understood as an attempt to make the field's inferential warrant match the strength of its claims. Where a study is exploratory, invisible multiplicity is the enemy; the remedy is to label exploratory analyses as such and to confirm them in an independent sample.



Worked example: evaluating a workplace stair-use prompt

Suppose a public health team wants to know whether point-of-choice prompts ("Take the stairs") increase stair use in an office building. Several decisions follow from the logic above.

- **Question type.** The question is causal and pragmatic: does the prompt change behaviour under real conditions? This is an effectiveness rather than efficacy question.
- **Operationalisation.** The primary outcome is stair entries per hundred building entrants per day, recorded by infrared counters on stairwell doors, validated against a fortnight of direct observation to estimate counter error.
- **Design.** Individual randomisation is impossible and would contaminate (colleagues share lifts and stairwells). A stepped-wedge cluster design — randomising twelve buildings into four waves of implementation — preserves randomised inference while accommodating staged rollout.
- **Threats.** Seasonality affects stair use; occupancy changes with holidays and remote-work policy; counters drift; participants may react to novelty. Calendar time is modelled as a fixed effect, and multiple pre-intervention measurement waves estimate baseline trend.
- **Analysis.** A multilevel model with random intercepts for building and observation day, pre-specified primary outcome, and an a priori power calculation based on the expected intraclass correlation.
- **Ethics and interpretation.** Because the intervention is environmental rather than individual, consent is usually waived with signage and an opt-out for data collection. Anticipated effects are small — a few

percentage points — but the intervention is nearly free, so the relevant question becomes cost-effectiveness rather than statistical significance in isolation.

Worked example: a single-case evaluation in a clinic

A practitioner wishes to evaluate a token-economy procedure for a child's on-task behaviour. A multiple-baseline design across three classroom settings, with staggered onset of the intervention over successive weeks, allows the inference that on-task behaviour increased when — and only when — the procedure was introduced in each setting. Repeated measurement, stable baselines, and visual plus statistical analysis of level and trend supply the warrant that a single before-and-after comparison cannot.

Ethics of Behavioural Research

Behavioural research raises ethical issues beyond the familiar requirements of review-board approval, informed consent, and confidentiality. Reactivity means that concealment is sometimes necessary for valid measurement, which puts deception and debriefing in tension with autonomy. Digital trace research complicates consent because data are often generated incidentally, are re-identifiable, and are held by third parties. Applied research conducted in organisations, clinics, and platforms blurs the boundary between research and service improvement, raising the question of when evaluation requires consent and when it constitutes ordinary practice. Finally, and most pointedly at the applied end of the field, interventions that shape behaviour without the actor's awareness — the design choices that behavioural scientists study and increasingly engineer — raise questions of transparency, manipulation, and consent that the field now treats as part of its subject matter rather than as external constraints.

From Evidence to Application

The applied reach of behavioural science has expanded sharply since the 2000s, through behavioural insights units in governments, the growth of behavioural finance and marketing, the design of digital choice architectures, and public health campaigns. The movement from evidence to practice follows a recognisable sequence: basic mechanism research identifies a lever; efficacy trials test it under controlled conditions; effectiveness trials test it in representative settings and populations; implementation research examines whether it was delivered as intended (fidelity), whether it was adapted appropriately, and whether it was sustained; and scale-up embeds it in routine practice, where it frequently encounters constraints that no trial anticipated.

Several regularities should temper expectations. Effect sizes typically shrink as interventions move from laboratory to field and from efficacy to effectiveness, because control tightens in the lab and loosens in practice. Interventions that work when delivered by enthusiasts may fail under routine staffing. Context — institutional, cultural, economic — moderates effects, which means that transportability rather than universality is the realistic goal, and that local evaluation is often indispensable. Against these cautions, the applied methods toolkit has widened considerably: pragmatic and adaptive trials, A/B testing at platform scale, n-of-1 trials for personalisation, and practice-based evidence generated within service systems rather than imported into them.

The methodological implications of application are not merely practical. Applied settings supply the phenomena that motivate basic research — the field is as often pulled by problems as pushed by theory — and they impose the constraint that a finding which cannot survive outside the laboratory has a limited claim to be a finding about behaviour at all.

Summary

- Methods are constitutive of behavioural science rather than auxiliary to it: the field's scientific standing rests on operationalised constructs, systematic observation, quantification, and replicability.
- Behavioural constructs are operationalised along response dimensions — frequency, latency, duration, intensity, choice, topography — and measured by observation, instrumentation, self-report, physiological indices, or digital traces, each with characteristic biases.
- Reliability is necessary but not sufficient for validity, and reactivity is a distinctive problem, mitigated by unobtrusive measurement and habituation but constrained by ethics.
- Causal warrant is graded: randomisation is strongest, with quasi-experimental designs (interrupted time series, difference-in-differences, regression discontinuity) and longitudinal observational designs supplying progressively weaker but often indispensable alternatives.
- Single-case experimental designs provide individual-level causal inference and remain central to applied and clinical work.
- Threats to internal validity — history, maturation, testing, instrumentation, regression, selection, attrition — have standard design countermeasures, and the replication crisis has driven pre-registration, open data, multi-site replication, and explicit effect-size reporting.
- Applied translation runs from mechanism through efficacy, effectiveness, implementation, and scale-up, with effect sizes generally attenuating as control is traded for realism.
- Ethical questions — deception, digital trace privacy, the research–practice boundary, and the transparency of behavioural intervention — are treated as internal to the field, not as external constraints on it.